

Why the Federal Reserve Cuts Rates when Public Debt Rises^{*}

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Abstract

We document that, conditional on inflation and output, the Federal Reserve lowers the policy rate when the U.S. public debt-to-GDP ratio rises. To explain this pattern, we develop and estimate a New Keynesian model with shocks to households' demand for public debt. These shocks generate a negative comovement between public debt and the natural rate. If monetary policy tracks the natural rate, the model rationalizes the negative relationship between debt and the policy rate. Consistent with this view, when a novel debt-informed natural rate measure is included in the policy rule, the debt-to-GDP ratio no longer explains variations in the policy rate.

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1 Introduction

Does the Federal Reserve respond to fiscal conditions, particularly the level of public debt, when setting interest rates? The prevailing view in macroeconomics holds that fiscal policy influences monetary decisions only indirectly, through its effects on inflation and output. Under this interpretation, public debt should not directly affect the conduct of monetary policy. Yet the Federal Reserve may face political pressure to lower interest rates when debt rises, easing the government's financing burden.

We contribute to this debate with a new fact: conditional on inflation and output, the Federal Reserve lowers the policy rate when the U.S. debt-to-GDP ratio rises. We argue that this behavior does not reflect political pressure. Instead, the Federal Reserve responds to the information that debt conveys about the natural rate of interest, the real interest rate that would prevail under flexible prices.

We support this interpretation by showing that shifts in households' demand for public debt explain the negative comovement between public debt and the policy rate. An increase in demand for government debt raises the quantity of debt and lowers its return, that is, the natural rate of interest. If the monetary authority targets the natural rate, these shifts imply a lower policy rate when public debt rises. Quantitatively, shifts in demand for safe assets account for most business cycle fluctuations. We construct a debt-informed measure of the natural rate and show that, once variation in this measure is taken into account, movements in public debt no longer affect the policy rate.

To establish that, conditional on inflation and output, the Federal Reserve lowers its policy rate when the U.S. debt-to-GDP ratio rises, we augment the standard Taylor rule¹ by including the debt-to-GDP ratio. We estimate the resulting equation using the generalized method of moments under alternative instrument sets. Across all specifications, the coefficient on debt is negative and statistically significant. We further show that this result cannot be explained by political pressure or by debt containing information about future inflation and output or financial conditions.

This empirical finding is difficult to reconcile with the textbook New Keynesian model

¹The standard Taylor rule relates the interest rate to inflation and the output gap, as in [Clarida et al. \(2000\)](#) and many others.

(Woodford, 2003; Galí, 2015). In that framework, increases in public debt result from higher deficits that raise output and inflation, inducing the monetary authority to increase the policy rate. The model therefore predicts a positive comovement between debt and interest rates.

To explain our empirical findings, we extend the textbook New Keynesian model in four ways. First, households derive utility from holding government bonds. As a result, fiscal policy plays a central role in shaping equilibrium outcomes and in determining the natural rate.² Second, we introduce shocks to the utility value of holding bonds that generate fluctuations in the demand for public debt. These shocks capture exogenous shifts in households' preference for holding safe and liquid assets, namely public debt, when the perceived value of safety rises or precautionary motives strengthen.³ Third, fiscal revenues respond not only to output and debt but also to the interest rate. Finally, monetary policy follows a Taylor rule that responds to inflation, the output gap, and the natural rate.

In the model, a positive shock to households' demand for public debt increases the demand for government bonds. The resulting increase in demand lowers the cost of debt issuance, inducing the fiscal authority to issue more bonds and households to increase saving, thereby reducing consumption. As a consequence, output and the natural rate decline. Because the monetary authority sets the policy rate in line with the natural rate, it lowers the nominal interest rate. This mechanism explains why the Federal Reserve reduces its policy rate when public debt rises: it responds not to debt itself but to the information it conveys about the natural rate.

The model also allows us to analyze shocks to the supply of public debt, such as an exogenous variation in taxes. In contrast to shocks to households' demand for public debt, shocks to the supply of public debt generate a positive comovement between the policy rate and public debt, consistent with the textbook New Keynesian model. Intuitively, an increase in the supply of debt raises both the quantity of debt and its return, the natural rate. As a result, the monetary authority increases the policy rate.

²In the standard New Keynesian model and its extensions, this mechanism is absent. As a result, the natural rate of interest is entirely independent of the level of public debt.

³For example, precautionary saving motives may increase when labor market conditions become more uncertain or when the economy is perceived as riskier.

The main insight of the model is that movements in public debt are informative about the natural rate. Monetary policy therefore does not respond to debt directly but to the information it contains about the natural rate. The model therefore implies that, once variation in the natural rate is taken into account, movements in public debt no longer affect the policy rate. We then test this implication empirically.

To this end, we construct a debt-informed measure of the natural rate, as existing estimates in the literature do not incorporate information about public debt.⁴ When this measure is included in our baseline Taylor rule specification, the coefficient on the debt-to-GDP ratio becomes insignificant, while the coefficient on the natural rate is positive and statistically significant. Excluding debt from the specification leaves the estimated coefficient on the natural rate virtually unchanged. These results support the view that debt affects the policy rate only through its information about the natural rate. However, they do not explain why public debt and the policy rate comove negatively.

Our theoretical framework implies that shocks to households' demand for public debt generate a negative comovement between public debt and the policy rate, whereas shocks to the supply of public debt generate a positive comovement. It is therefore necessary to assess which type of shock accounts for business cycle fluctuations in public debt, the natural rate, and the policy rate. To this end, we develop and estimate a New Keynesian DSGE model that extends [Justiniano et al. \(2010\)](#)'s framework by incorporating the elements described above. In the model, households derive utility from holding government bonds, fiscal revenues respond to output, debt, and the interest rate, and monetary policy follows a Taylor rule that responds to inflation, the output gap, and the natural rate.

We estimate the model using standard Bayesian methods. The variance decomposition indicates that shocks to households' demand for public debt account for roughly 40 percent of the fluctuations in output, public debt, the policy rate, and the natural rate. Because these shocks move both public debt and the natural rate, public debt is informative about movements in the natural rate. In particular, following such a shock, public debt comoves negatively with both the natural rate and the policy rate. Moreover, this shock generates

⁴[Laubach and Williams \(2003\)](#) note that fiscal policy may affect the natural rate but do not model this channel explicitly.

positive comovement among consumption, investment, output, inflation, and the policy rate, as documented for aggregate demand shocks in the data (Angeletos et al., 2020). These dynamics help explain why these shocks are a major driver of business cycle fluctuations.

To show that the model can replicate our empirical findings, we conduct a simulation exercise. We generate data from the estimated model under a Taylor rule that responds to the natural rate but not to public debt. Using these simulated data, we then estimate a Taylor rule that includes public debt instead of the natural rate, as in our empirical specification. The estimated coefficient on public debt is negative, consistent with our empirical results and arising from the tight link between public debt and the natural rate in the model.

In contrast, when we repeat the simulation exercise allowing only for shocks to the supply of public debt, namely tax and government spending shocks, the estimated coefficient on public debt becomes positive and statistically significant. This result reflects the positive comovement between public debt and the natural rate generated by these shocks.

In sum, our empirical finding that the Federal Reserve lowers the policy rate when public debt rises does not reflect fiscal accommodation. Instead, public debt is informative about movements in the natural rate of interest, and monetary policy responds to the natural rate rather than to debt itself. In our sample, increases in public debt are primarily driven by shocks to households' demand for public debt, which lower the natural rate and, in turn, the policy rate. Had shocks to the supply of public debt been dominant, public debt and the policy rate would have comoved positively.

Related literature. We make several contributions to the literature. First, we extend the literature estimating the natural rate of interest (Laubach and Williams, 2003; Lubik and Matthes, 2015; Holston et al., 2017; Del Negro et al., 2017), which abstracts from public debt. We show that movements in public debt affect the natural rate and construct a debt-informed measure that incorporates this information.

Second, our work relates to studies of fiscal–monetary interactions (Sargent and Wallace, 1981; Leeper, 1991; Christiano et al., 2011; Leeper et al., 2017; Bianchi et al., 2023; Smets and Wouters, 2024). Most existing analyses focus on interactions driven by government spending shocks, whereas we emphasize shocks to households' demand for public debt. This perspective explains the negative comovement between nominal interest rates and public

debt.

Third, we build on the literature that embeds bonds in household utility (Campbell et al., 2017; Kaplan and Violante, 2018; Hagedorn, 2018; Michaillat and Saez, 2021). We depart from the standard assumption of zero net bond supply and instead allow bonds—interpreted as public debt—to fluctuate over the business cycle. We also introduce shocks to the utility of bonds, which capture shifts in the demand for safe assets and admit several interpretations. They microfound the “risk-premium” shock of Smets and Wouters (2007) when debt supply is zero (Fisher, 2015); appear as spread or flight-to-safety shocks in the international literature (Eichenbaum et al., 2021; Kekre and Lenel, 2024); and can be viewed as increases in idiosyncratic risk (Aiyagari, 1994; Bayer et al., 2019) or in the riskiness of capital returns (Li and Merkel, 2025). In each case, households shift resources toward government bonds and away from capital, reducing investment and consumption and generating a recession.

Fourth, our framework is closely connected to the literature on analytically tractable heterogeneous-agent New Keynesian (HANK) models. Kaplan and Violante (2018), Auclert et al. (2024), Auclert et al. (2025), and Wolf (2025) show that bond-in-utility models reproduce key aggregate behavior of HANK economies while remaining tractable.⁵ Our approach is particularly close to Angeletos et al. (2024a,b), who construct HANK models without explicit heterogeneity. In their setting, debt affects household behavior because agents are finitely lived; in ours, because bonds provide utility. We develop and estimate, rather than calibrate, a quantitative model that embeds this mechanism together with standard DSGE frictions.

Finally, our analysis is related to Summers and Rachel (2019), Campos et al. (2024), and Nuño (2025), who examine how variations in public debt affect the natural rate of interest. These papers study the long-run relationship between the natural rate and public debt, whereas we focus on their business cycle relationship. We contribute by showing that shocks to households’ demand for public debt generate negative comovement between public debt and the natural rate.

Outline. The paper is organized as follows. Section 2 documents our empirical find-

⁵Auclert et al. (2024, 2025) show that adding hand-to-mouth households improves the match to the intertemporal marginal propensity to consume out of labor income in HANK models.

ing that the Federal Reserve lowers the policy rate when public debt increases. [Section 3](#) presents a stylized New Keynesian model that explains this empirical finding. [Section 4](#) constructs a debt-informed measure of the natural rate and shows that it explains the observed relationship between debt and the policy rate. [Section 5](#) estimates and discusses the results of our quantitative New Keynesian DSGE model. [Section 6](#) concludes.

2 Empirical Strategy and Results

In this section, we show that the Federal Reserve adjusts its policy rate in response not only to inflation and output but also to the public debt-to-GDP ratio. We estimate a standard Taylor rule augmented with public debt to establish this result.

[Section 2.1](#) presents the benchmark specification. [Section 2.1.1](#) confirms the robustness of the result across alternative measures of inflation and different sample periods. [Section 2.1.2](#) addresses concerns about time trends by using complementary empirical strategies. Finally, [Section 2.2](#) rules out alternative explanations, such as the possibility that public debt contains information about future economic conditions.

2.1 Main empirical result

Following [Taylor \(1993\)](#), the Taylor rule provides a benchmark for how the policy rate responds to movements in inflation and the output gap. We augment this specification by including the public debt-to-GDP ratio:

$$r_t = c + \alpha\pi_t + \beta y_t + \gamma d_t + \rho_1 r_{t-1} + \rho_2 r_{t-2} + u_t \quad (1)$$

where r_t is the yearly federal funds rate, π_t is inflation, y_t is the output gap, d_t is the public debt-to-GDP ratio, and u_t is the error term, interpreted as the monetary policy shock. We focus on estimating the coefficient γ , which captures the response of the policy rate to movements in public debt: a negative value indicates that the Federal Reserve lowers the policy rate when the debt-to-GDP ratio rises. For interpretability, we rescale d_t so that γ measures the effect of a ten-percentage-point increase in the debt-to-GDP ratio.

A challenge in estimating [equation \(1\)](#) is potential endogeneity: inflation, the output gap and debt may be correlated with the residual, interpreted as the monetary policy shock. We first estimate [equation \(1\)](#) by ordinary least squares (OLS). Monetary policy shocks account for only a small share of business-cycle fluctuations;⁶ it follows that any bias resulting from OLS estimation is limited ([Carvalho et al., 2021](#)). We then employ two instrumental-variable strategies. Following [Clarida et al. \(2000\)](#), we estimate the equation by generalized method of moments (GMM) using lagged variables as instruments, which are predetermined relative to policy shocks. Following [Barnichon and Mesters \(2020\)](#), we also use macroeconomic shocks orthogonal to monetary policy shocks as instruments.

We estimate [equation \(1\)](#) using quarterly data from 1979:Q3 to 2019:Q4. [Table 1](#) reports the results across different estimation procedures. Column (1) presents OLS estimates. Column (2) reports GMM estimates using the same instruments as in [Clarida et al. \(2000\)](#). Column (3) uses GMM with lagged variables and the business-cycle shock of [Angeletos et al. \(2020\)](#).⁷ Column (4) replaces this shock with the excess bond-premium shock of [Gilchrist and Zakrajšek \(2012\)](#).⁸ Column (5) employs only lagged macroeconomic shocks as instruments.⁹ All coefficients are divided by $1 - \rho \equiv 1 - \rho_1 - \rho_2$, so that the reported parameters correspond to $\pi_t^{cpi} \equiv \frac{\alpha}{1-\rho}$, $y_t^* \equiv \frac{\beta}{1-\rho}$, $d_t \equiv \frac{\gamma}{1-\rho}$, and $\rho \equiv \rho_1 + \rho_2$. This scaling follows standard practice in the literature and allows interpretation of the coefficients as long-run effects.

⁶This is well documented in the DSGE literature; see [Smets and Wouters \(2007\)](#), [Justiniano et al. \(2010\)](#), [Justiniano et al. \(2011\)](#), [Christiano et al. \(2014\)](#), and [Smets and Wouters \(2024\)](#).

⁷The shock of [Angeletos et al. \(2020\)](#) is not perfectly orthogonal to monetary policy shocks, since it is a linear combination of all structural shocks. However, as noted by [Carvalho et al. \(2021\)](#), monetary policy shocks explain little of business-cycle variation and therefore receive negligible weight in this linear combination.

⁸This shock is identified using a Cholesky decomposition in a VAR.

⁹Further details on the data and estimation procedure are reported in [Appendix A](#) and [Appendix B](#), respectively.

Table 1: Taylor rule with public debt-to-GDP ratio

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
π_t^{cpi}	1.17*** (0.25)	1.11*** (0.22)	1.30*** (0.17)	1.20*** (0.22)	1.18*** (0.12)
y_t	0.60*** (0.19)	0.73*** (0.17)	0.84*** (0.16)	0.83*** (0.16)	0.45*** (0.11)
d_t	-0.74*** (0.22)	-0.80*** (0.18)	-0.55*** (0.16)	-0.59*** (0.18)	-0.95*** (0.10)
ρ	0.68*** (0.11)	0.76*** (0.05)	0.72*** (0.03)	0.78*** (0.04)	0.63*** (0.04)
N	162	159	162	162	144

Notes: The table reports estimates of [equation \(1\)](#). The dependent variable is the federal funds rate, r_t . The explanatory variables are CPI inflation, π_t , the output gap, y_t , and the public debt-to-GDP ratio, d_t . The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. The variable d_t is rescaled so that one unit corresponds to 10 percentage points. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of [Clarida et al. \(2000\)](#), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of [Angeletos et al. \(2020\)](#) and the excess bond-premium shock of [Gilchrist and Zakrajšek \(2012\)](#). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of [Ramey \(2016\)](#), and the news oil-price shocks of [Känzig \(2021\)](#). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

[Table 1](#) reveals a consistent pattern across all specifications: the coefficient on the debt-to-GDP ratio is negative and statistically significant. On average, a ten-percentage-point increase in the debt-to-GDP ratio lowers the federal funds rate by about 70 basis points, conditional on inflation and the output gap. The estimated coefficients are also remarkably similar across columns, suggesting that endogeneity is limited and that public debt helps explain interest rate decisions. Consistent with the literature, the coefficients on inflation and the output gap are positive and statistically significant. Their magnitudes are in line with standard Taylor rule estimates, with the inflation coefficient around 1.5 and the output-gap coefficient around 0.6.

2.1.1 Robustness of the main specification

In this section, we assess the robustness of the results reported in [Table 1](#). All tables discussed below are constructed in the same way as [Table 1](#).

The results are not sensitive to measuring inflation using the GDP deflator or the PCE price index ([Table B1](#) and [Table B2](#), respectively), to using the federal funds rate as the policy variable instead of the shadow rate of [Wu and Xia \(2016\)](#) ([Table B3](#)), and to excluding the zero lower bound period by ending the sample in 2011:Q3 ([Table B4](#)).

A possible concern is that, in our baseline specification, we divide public debt at time t by GDP at time t , and the estimated coefficient might capture contemporaneous movements in GDP rather than debt. [Table B5](#) shows that the results remain robust if we divide public debt by GDP at time $t - 1$. Similarly, [Table B6](#) reports estimates obtained when the debt-to-GDP ratio is constructed using potential GDP at time t instead of actual GDP, leading to the same conclusion.

Finally, following [Clarida et al. \(2000\)](#), we estimate the Taylor rule using realized variables at time $t + 1$, under the assumption that the Federal Reserve is forward looking and responds to expected rather than contemporaneous conditions. Accordingly, we replace inflation and the output gap in [equation \(1\)](#) with their realized values at time $t + 1$. [Table B7](#) confirms that the coefficient on the debt-to-GDP ratio remains negative and statistically significant.

In summary, across all robustness exercises, the estimated coefficient on public debt remains negative, statistically significant, and similar in magnitude to the benchmark specification. These results confirm our finding that the Federal Reserve lowers the policy rate when the debt-to-GDP ratio rises.

2.1.2 Controlling for low-frequency variation

Another potential concern is that our results may be driven by low-frequency comovements in debt, inflation, output, and interest rates rather than higher-frequency fluctuations.

As noted by [Granger and Newbold \(1974\)](#), if this were the case, the regression residuals should exhibit serial correlation. To test for this possibility, [Table B8](#) reports the results of the [Cumby and Huizinga \(1992\)](#) serial correlation test. Each column corresponds to the

respective specification in [Table 1](#), and each row reports the test statistic under the null that all autocorrelation coefficients up to the given lag are zero. We find no evidence of serial correlation in the residuals, suggesting that our results are not driven by low-frequency trends.

We further address time-trend concerns in two ways. First, we estimate the Taylor rule using detrended series for inflation, output, interest rates, and the debt-to-GDP ratio.¹⁰ [Table B9](#) reports the results and confirms our main finding: the coefficient on the debt-to-GDP ratio is negative and statistically significant, indicating that debt has an economically meaningful effect on monetary policy decisions.

The second method to address low-frequency concerns employs the system-projections instrumental-variables (SP-IV) method of [Lewis and Mertens \(2022\)](#), which estimates the Taylor rule in the impulse-response-function space, where variables do not exhibit trends.¹¹ [Table B10](#) reports the results, showing once more that the coefficient on the debt-to-GDP ratio is negative and statistically significant.

Taken together, these results indicate that the relationship we document reflects policy behavior rather than spurious comovement due to common trends.

2.2 Ruling out alternative explanations

Taylor rules provide a parsimonious description of how monetary policy responds to macroeconomic conditions. In practice, however, the Federal Reserve bases its policy decisions on a broader information set than realized inflation and the output gap. It is therefore possible that the estimated relationship between the policy rate and the debt-to-GDP ratio reflects other factors correlated with debt that the Federal Reserve considers when setting interest rates. In this section, we empirically examine and rule out several such explanations.

The main idea is that if another variable influences interest-rate decisions through its effect on the public debt-to-GDP ratio, then controlling for that variable should eliminate the significance and explanatory power of debt. To test this hypothesis, we sequentially

¹⁰See [Appendix B](#) for details on the filtered techniques used.

¹¹Intuitively, the SP-IV method can be viewed as a two-step procedure. In the first step, impulse responses at different horizons are estimated for each variable. In the second step, the Taylor rule is estimated using these responses in place of the original variables.

add potential control variables to the Taylor rule. We therefore estimate the following specification:

$$r_t = c + \alpha\pi_t + \beta y_t + \gamma d_t + \rho_1 r_{t-1} + \rho_2 r_{t-2} + \phi x_t + u_t, \quad (2)$$

where all variables are defined as before, and x_t denotes the additional control variables. In the following paragraphs, we examine different possible mechanisms.

2.2.1 Public debt as a predictor of future inflation or the output gap

It is well established that the Federal Reserve conducts monetary policy in a forward-looking manner, responding not only to current inflation and output but also to their expected future values (Clarida et al., 2000). Accordingly, if the debt-to-GDP ratio helps predict future inflation or output, its significance in the estimated Taylor rule could simply reflect this forward-looking behavior. To assess this possibility, we proceed in several ways.

First, we estimate equation (1) using the average of inflation and the output gap over the next four quarters as regressors.¹² Table B11 reports the results, showing that the coefficient on the debt-to-GDP ratio remains negative and statistically significant. Its magnitude is also close to our benchmark specification, with an effect of about 50 basis points for a ten-percentage-point increase in public debt. We obtain the same conclusion when replacing realized values with Greenbook forecasts, averaged over the same four future quarters, as reported in Table B12. In both Tables B11 and B12, the coefficients on inflation and the output gap are positive and statistically significant.

Second, we estimate equation (2) by adding as controls the realized future values of inflation and the output gap over horizons of three, five, and ten years. Tables B13, B14, and B15 report the results, showing the same pattern as before.

These robustness exercises indicate that the negative relationship between the policy rate and the public debt-to-GDP ratio is not driven by information about future inflation or output. Public debt therefore appears to influence interest-rate decisions through a different channel.

¹²This approach—using realized future values averaged over several quarters—is common in empirical macroeconomics; see, for example, Barnichon and Mesters (2020).

2.2.2 Political pressure

Another possible explanation for our findings is that the Federal Reserve lowers the policy rate when the public debt-to-GDP ratio rises because of political pressure from the government. To examine this possibility, we add, as separate controls, the number of interactions—phone calls and in-person meetings—between the U.S. president and the Federal Reserve chair, taken from Drechsel (2024), and a measure of proximity to elections.¹³ We therefore estimate equation (2) controlling for each of these two measures of political pressure in turn. Tables B16 and B17 show that the coefficient on the public debt-to-GDP ratio remains negative and statistically significant, with a magnitude similar to that in our benchmark specification, Table 1. We interpret these findings as evidence that public debt influences monetary policy decisions beyond the direct effects of political pressure.

2.2.3 Economic conditions

A related concern is that the public debt-to-GDP ratio may reflect broader economic conditions. For example, during recessions, public debt typically rises as government spending increases, while the Federal Reserve lowers interest rates to stimulate the economy. It is therefore possible that the level of public debt simply reflects the state of the business cycle. To test this hypothesis, we estimate equation (2) including an indicator for NBER recession dates as an additional control (Table B18). The recession indicator is negative and significant, consistent with the view that interest rates decline during downturns. However, the debt-to-GDP ratio continues to affect monetary policy beyond economic downturns, confirming that our results are robust to controlling for recessions.

Another potential explanation is that financial conditions are correlated with the demand for public debt, so that the debt-to-GDP ratio merely captures how easily households and firms can borrow. Indeed, Caballero et al. (2024) show that the Federal Reserve lowers interest rates when financial conditions deteriorate. To account for this possibility, we add as controls in equation (2) the BAA–AAA corporate bond spread, the Gilchrist and Zakrajšek (2012) spread, or the Chicago Fed National Financial Conditions Index (Tables B19, B20,

¹³This variable is defined as the number of quarters until the next midterm or presidential election.

and B21). We also control for stock market conditions using either the CBOE Volatility Index (VIX) or S&P 500 returns (Tables B22 and B23). All these exercises show largely unaltered results, suggesting that public debt influences monetary policy independently of financial conditions or movements in equity markets.

2.3 Summary

Across all specifications and robustness exercises, we find a consistent result: when the public debt-to-GDP ratio rises, the Federal Reserve lowers the policy rate, conditional on inflation and the output gap. The effect is quantitatively stable and statistically significant across estimation methods, alternative measures of inflation, policy rates, and detrending approaches. It also remains robust when controlling for recessions, political pressure, financial and stock-market conditions. This evidence indicates that public debt contains independent information relevant for monetary policy decisions.

This empirical regularity raises a natural question: through which mechanism does public debt influence interest-rate setting? The next section develops a theoretical framework to rationalize this finding.

3 A stylized New Keynesian model

This section presents a stylized New Keynesian model to explain our empirical finding that the Federal Reserve lowers the policy rate when the public debt-to-GDP ratio rises. The framework builds on the standard three-equation New Keynesian model, augmented with bonds in households' utility, shocks to the utility value of holding bonds, and a fiscal rule in which taxes respond to debt, the interest rate, and output.

The key insight of the model is that shocks to households' demand for public debt generate negative comovements between the policy rate and public debt. These shocks lower the natural rate of interest, prompting a monetary authority that targets the natural rate to reduce the policy rate. Hence, the monetary authority does not respond to debt directly, but to the information that debt conveys about the natural rate.

3.1 Linearized model

In this section, we present the model's equilibrium conditions linearized around a steady state with zero inflation. A detailed derivation is provided in [Appendix C](#).

$$y_t = \mathbb{E}_t y_{t+1} - (r_t - \mathbb{E}_t \pi_{t+1}) + \phi_B d_t - \eta_t \quad (\text{E.E.})$$

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa y_t \quad (\text{NKPC})$$

$$r_t = \phi_{R^*} R_t^* + \phi_\pi \pi_t + \phi_y y_t \quad (\text{TR})$$

$$d_t = R(r_{t-1} + d_{t-1} - \pi_t) - \frac{tax}{d} tax_t \quad (\text{gov't b.c.})$$

$$tax_t = \tau_y y_t + \tau_d d_{t-1} + \tau_R(r_t - \mathbb{E}_t \pi_{t+1}) + \tau_t \quad (\text{tax rule})$$

Here, r_t denotes the nominal interest rate, y_t the output gap, π_t inflation, d_t the real value of public debt, tax_t lump-sum taxes in real terms, and R_t^* the natural rate of interest—that is, the equilibrium real rate in the flexible-price economy. The shocks η_t and τ_t follow AR(1) processes, and variables without time subscripts represent steady-state values.

[Equation \(E.E.\)](#) is the Euler equation and features the novel term $\phi_B d_t$, which arises from the assumption that government bonds enter households' utility. Under this assumption, output, and therefore consumption, depends directly on the level of public debt, breaking Ricardian equivalence. It also delivers a tractable framework that replicates key features of heterogeneous-agent models without requiring us to track the distribution of agents.¹⁴ The parameter ϕ_B governs the strength of this channel. When $\phi_B = 0$, bonds provide no utility services, and the Euler equation reverts to its standard textbook form.

In [equation \(E.E.\)](#), the shock η_t represents changes in the utility value of holding government bonds. This shock captures exogenous shifts in households' preference for safe and liquid assets—public debt—when the perceived value of safety rises or precautionary motives strengthen. More broadly, η_t can be interpreted as a liquidity or safety shock that reflects increases in precautionary saving behavior during periods of greater uncertainty or perceived

¹⁴Models that replicate key features of heterogeneous-agent frameworks without tracking the distribution of agents are often referred to in the literature as analytically tractable heterogeneous-agent New Keynesian models ([Kaplan and Violante, 2018](#); [Auclert et al., 2025](#); [Angeletos et al., 2024b](#)).

risk—such as when labor-income prospects weaken or financial conditions tighten. In this linearized framework, η_t is equivalent to a discount-factor shock, although the two differ in the quantitative model presented in [Section 5](#).

[Equation \(NKPC\)](#) is the standard New Keynesian Phillips curve implied by price rigidities. [Equation \(TR\)](#) specifies the Taylor rule, which differs from the conventional formulation through the term $\phi_{R^*} R_t^*$: the monetary authority adjusts the nominal rate in response to movements in the natural rate. This assumption is central to our mechanism and provides a theoretical foundation for the empirical finding in [Section 2](#): public debt is informative about the natural rate, but the monetary authority responds only to the natural rate itself.

[Equation \(gov't b.c.\)](#) is the linearized government budget constraint in real terms, and [equation \(tax rule\)](#) specifies the fiscal rule. The parameter τ_y captures the responsiveness of taxes to output fluctuations, τ_d governs how actively the government stabilizes debt, and τ_R measures the sensitivity of taxes to the real interest rate. The presence of τ_R is novel: it reflects that higher interest rates raise the cost of servicing public debt, inducing the fiscal authority to increase tax revenues. Finally, the shock τ_t represents an exogenous tax disturbance, capturing unexpected shifts in fiscal policy. A negative realization of τ_t can be interpreted as a fiscal stimulus or a transfer to households.

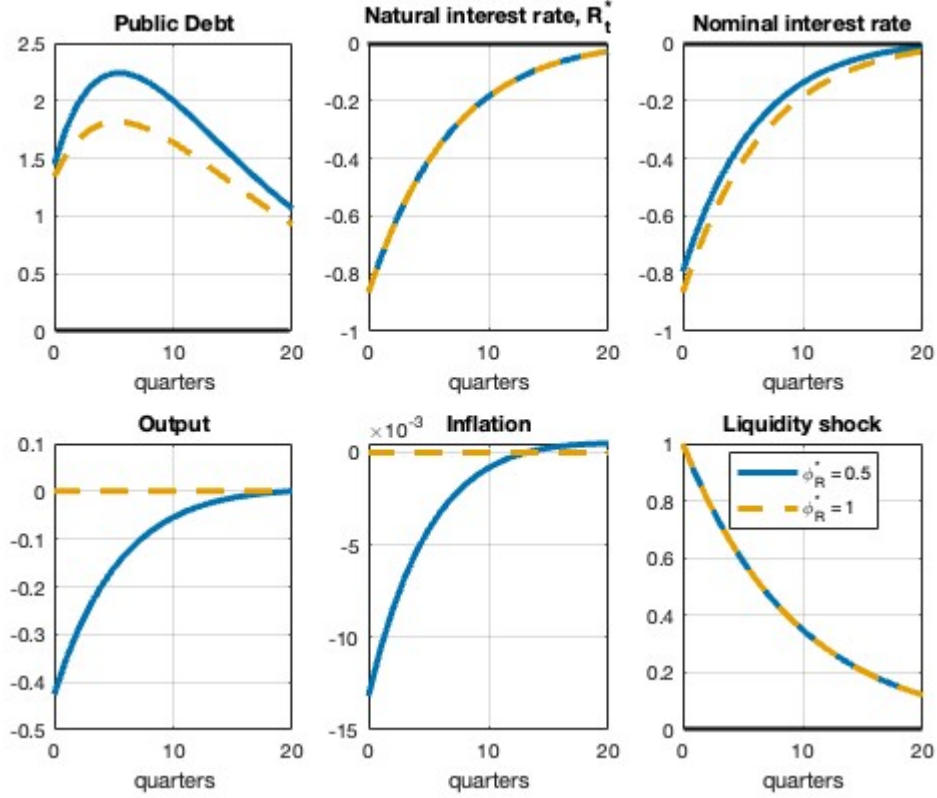
We next illustrate the dynamic implications of the model by examining, in turn, the responses of key variables to a liquidity shock and to a tax shock.

3.2 Impulse response functions

[Figure 1](#) reports the impulse response functions (IRFs) to a shock to households' demand for public debt, that is, a liquidity shock. The panels in the figure display, respectively, the responses of public debt, the natural rate, the nominal interest rate, output, inflation, and the liquidity shock, under two alternative Taylor rules. In the first rule, the coefficient on the natural rate is set to $\phi_{R^*} = 0.5$ (blue line); in the second, to $\phi_{R^*} = 1$ (yellow line). When $\phi_{R^*} = 1$, the monetary authority fully stabilizes the economy: output and inflation remain at their steady-state levels, replicating the flexible-price allocation for these variables. This case corresponds to the familiar *divine coincidence* of the textbook New Keynesian model, in which monetary policy can simultaneously stabilize inflation and the output gap in response

to demand shocks.

Figure 1: Impulse response function to a liquidity shock



Notes: The figure shows the impulse response functions to a one-standard-deviation liquidity (safety) shock that raises households' preference for holding government bonds. The blue line corresponds to the Taylor rule with $\phi_{R^*} = 0.5$, while the yellow line corresponds to the rule with $\phi_{R^*} = 1$. The panels display, in order, the responses of public debt, the natural rate of interest, the nominal interest rate, output, inflation, and the liquidity shock itself. The figure highlights the negative comovement between public debt and the natural rate implied by the model's liquidity shock.

In **Figure 1**, debt and the nominal interest rate exhibit a negative comovement under both Taylor rules. Indeed, a positive liquidity shock increases households' demand for safe and liquid assets, leading them to reduce consumption and increase saving. In the case with $\phi_{R^*} = 0.5$ (blue line), this behavior lowers output and generates deflation, since the nominal interest rate does not decrease enough to offset the contraction in goods demand. When $\phi_{R^*} = 1$ (yellow line), monetary policy adjusts one-for-one with movements in the natural rate, preventing fluctuations in output and inflation. The reallocation of resources induced by the liquidity shock lowers the natural rate, R_t^* . As the central bank tracks movements

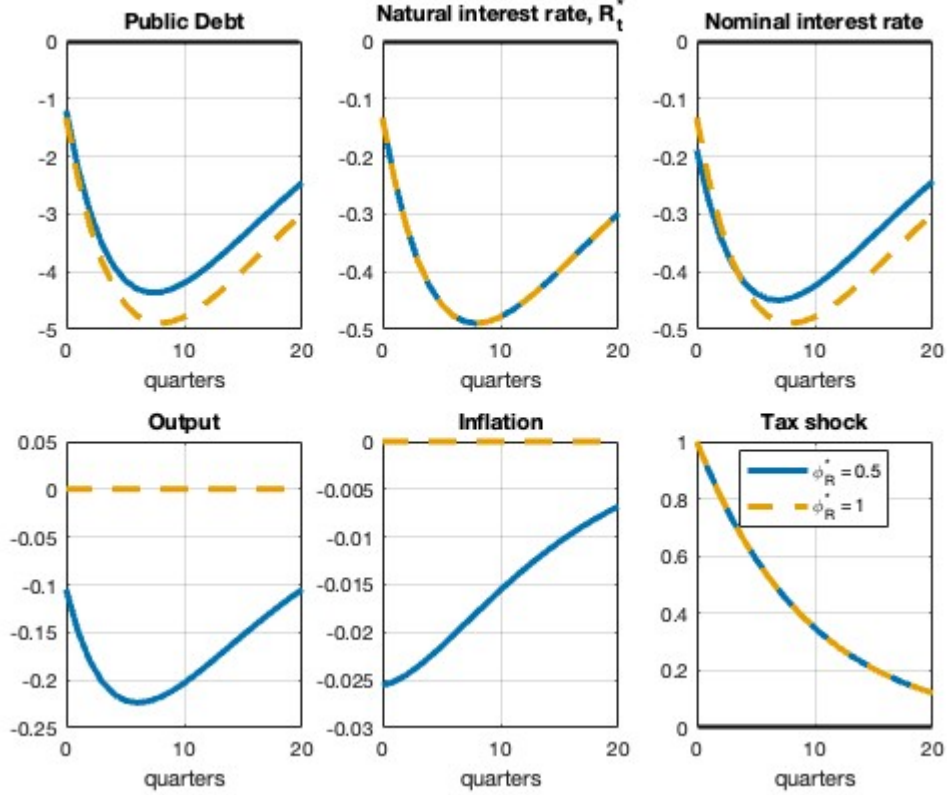
in the natural rate, it reduces the nominal policy rate accordingly. Hence, even though the monetary authority does not respond to debt directly, debt remains highly informative about the natural rate. When public debt rises because of a stronger household demand for safe assets—a liquidity shock—the natural rate declines, and the central bank lowers the nominal interest rate.

The fiscal rule plays a key role in sustaining this mechanism. When taxes respond strongly to interest-rate movements—that is, when τ_R is large (close to one) in [equation \(11\)](#)—the fiscal authority accommodates changes in households’ demand for debt, allowing debt issuance to adjust smoothly. When τ_R is small (for instance, $\tau_R = 0.2$), the government’s debt supply becomes more inelastic, and the negative comovement between public debt and the natural rate disappears. In this case, the fiscal authority does not expand debt even when the cost of borrowing declines, lowering taxes. [Figure C.1](#) confirms that, when $\tau_R = 0.2$, the correlation between debt and the natural rate becomes positive. In the quantitative model estimated in [Section 5](#), we find that τ_R is indeed close to one and statistically significant, consistent with this interpretation. The corresponding estimates are reported in [Table F.1](#).

Our analysis suggests that demand-driven changes in public debt (liquidity shocks) generate a negative comovement between public debt and the nominal interest rate. The model also allows us to examine how this comovement changes when debt is driven by supply-side disturbances, such as a tax shock that increases fiscal revenues.

To illustrate this case, [Figure 2](#) reports the impulse responses to a tax shock, constructed analogously to [Figure 1](#), except that the last panel now displays the tax shock instead of the liquidity shock. The qualitative effects of a tax shock are independent of the monetary policy rule. When the fiscal authority raises taxes, public debt declines and households’ wealth falls. Feeling poorer, households reduce consumption, generating a recession. To mitigate this downturn, the monetary authority lowers the policy rate. Tax shocks therefore induce a *positive* comovement between the nominal interest rate and public debt. The same mechanism operates in the flexible-price economy: higher taxes reduce wealth and consumption, lowering output and the natural rate of interest. Hence, under the assumption that the central bank tracks the natural rate, tax shocks naturally produce a positive association

Figure 2: Impulse response functions to a tax shock



Notes: The figure shows the impulse response functions to a one-standard-deviation tax shock that increases the primary surplus. The blue line corresponds to the Taylor rule with $\phi_{R^*} = 0.5$, while the yellow line corresponds to the rule with $\phi_{R^*} = 1$. The panels display, in order, the responses of public debt, the natural rate of interest, the nominal interest rate, output, inflation, and the tax shock itself. The figure highlights the positive comovement between public debt and the natural rate implied by the model's tax shock.

between public debt and the policy rate.¹⁵

The model can generate either positive or negative comovement between public debt and the natural rate, depending on the source of the disturbance. Liquidity (demand) shocks generate a negative correlation between public debt and the natural rate, whereas tax (supply) shocks produce a positive one.¹⁶ This distinction implies that our empirical finding of a negative coefficient on debt reflects the predominance of liquidity shocks over

¹⁵The assumption that bonds enter households' utility, i.e., $\phi_B \neq 0$, is crucial for tax shocks to affect the real economy. When $\phi_B = 0$, Ricardian equivalence holds and fiscal shocks have no effect on output, inflation, or the interest rate. This implication is inconsistent with the established empirical evidence that tax changes affect output, see, among many others, (Mertens and Ravn, 2012).

¹⁶We classify liquidity shocks as demand shocks because they shift the demand curve for public debt. By contrast, tax shocks shift the supply curve for public debt, so we classify them as supply shocks, even though both shocks generate a positive comovement between inflation and output.

fiscal ones. Had tax shocks been the dominant driver of fluctuations, we would instead have found a positive relationship between debt and the policy rate. In [Section 5](#), we extend the analysis to a quantitative DSGE framework to assess which type of shock dominates in explaining business-cycle dynamics.

4 The Natural Interest Rate

The model presented in [Section 3](#) provides a mechanism consistent with our empirical finding that the Federal Reserve lowers the policy rate when public debt increases. The key idea is that the Federal Reserve does not target public debt directly but the natural rate of interest, and that public debt contains information about this measure. This implies that, once we control for the natural rate in [equation \(2\)](#), the coefficient on public debt should lose its significance.

To test this implication, we proceed as follows. In [Section 4.1](#), we include existing estimates of the natural rate from the literature as an additional control in [equation \(2\)](#). The coefficient on the public debt-to-GDP ratio remains negative and statistically significant. This result reflects the fact that the existing measures of the natural rate do not incorporate information about public debt.

In [Section 4.2](#), we therefore construct a measure of the natural rate that explicitly incorporates information on public debt. When this debt-augmented natural rate is added to [equation \(2\)](#), the coefficient on the public debt-to-GDP ratio becomes insignificant, while the coefficient on the natural rate is positive and statistically significant.

4.1 Estimates in the literature of the natural rate

Guided by the model’s intuition in [Section 3](#), we estimate [equation \(2\)](#) controlling for the measure of the natural rate of interest constructed by [Lubik and Matthes \(2015\)](#). [Table 2](#) reports the results and follows the same structure as [Table 1](#). Column (1) presents the OLS estimates, and the remaining columns report GMM estimates using alternative sets of instruments. In all specifications, the coefficient on the public debt-to-GDP ratio remains negative and statistically significant. Its magnitude, however, is roughly half of that reported

in Table 1, consistent with the idea that debt is correlated with and informative about the natural rate. Since the Lubik and Matthes (2015) measure does not incorporate information on public debt, it cannot fully capture the underlying natural rate. While the coefficient on inflation is positive and statistically significant, as in Table 1, the coefficient on the output gap is negative and generally insignificant. This pattern reflects that the output gap and the natural rate measure convey similar information, making it difficult to disentangle their effects empirically.

Table 2: Taylor rule using the natural rate estimated by Lubik and Matthes (2015)

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
π_t	1.15*** (0.11)	1.19*** (0.08)	1.23*** (0.06)	1.20*** (0.08)	1.02*** (0.05)
y_t	-0.14 (0.12)	-0.07 (0.08)	-0.11* (0.06)	-0.07 (0.08)	-0.36*** (0.05)
d_t	-0.28** (0.12)	-0.34*** (0.08)	-0.25*** (0.07)	-0.29*** (0.08)	-0.20*** (0.06)
$R_t^{*,lm}$	4.41*** (0.63)	3.66*** (0.40)	4.12*** (0.37)	3.84*** (0.35)	5.14*** (0.31)
ρ	0.47*** (0.13)	0.48*** (0.05)	0.37*** (0.05)	0.49*** (0.05)	0.41*** (0.03)
N	162	159	162	162	144

Notes: The table reports estimates of equation (2). The dependent variable is the federal funds rate, r_t . The explanatory variables are CPI inflation, π_t , the output gap, y_t , the public debt-to-GDP ratio, d_t , and the natural rate of interest estimated by Lubik and Matthes (2015), $R_t^{*,lm}$. The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. The variable d_t is rescaled so that one unit corresponds to 10 percentage points. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of Clarida et al. (2000), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of Angeletos et al. (2020) and the excess bond-premium shock of Gilchrist and Zakrajšek (2012). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of Ramey (2016), and the news oil-price shocks of Känzig (2021). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

These results are not specific to the Lubik and Matthes (2015) measure. When we instead use alternative estimates of the natural rate from Laubach and Williams (2003) and Holston et al. (2017), the coefficient on the public debt-to-GDP ratio remains negative and

statistically significant. [Table D.1](#) and [Table D.2](#) report the corresponding results from estimating [equation \(2\)](#) with these measures.¹⁷ In all these cases, the coefficients on inflation and the output gap are positive, while the coefficient on the natural rate is negative and often insignificant. This likely reflects the fact that the existing measures of the natural rate—including [Lubik and Matthes \(2015\)](#), [Laubach and Williams \(2003\)](#), and [Holston et al. \(2017\)](#)—are constructed in a similar way, relying on information closely tied to output. In particular, they typically assume that the natural rate depends primarily on trend productivity growth, which is highly correlated with the output gap. As a result, either the natural rate or the output gap tends to be significant, but not both.

However, other factors—such as the demand for public debt—also affect the natural rate, as in the model described in [Section 3](#). For this reason, it is important to construct a measure of the natural rate that incorporates information on public debt.

4.2 New measure of the natural rate incorporating public debt

In the existing literature, measures of the natural rate are typically constructed using information on output growth, inflation, and the real interest rate ([Laubach and Williams, 2003](#); [Lubik and Matthes, 2015](#); [Holston et al., 2017](#)). In what follows, we modify this estimation by incorporating the public debt-to-GDP ratio.

Following [Lubik and Matthes \(2015\)](#), we estimate the natural rate using a time-varying parameter vector autoregression (TVP-VAR). The model, originally developed by [Primiceri \(2005\)](#), can be written as

$$y_t = c_t + B_{1,t}y_{t-1} + \cdots + B_{k,t}y_{t-k} + u_t, \quad t = 1, \dots, T, \quad (3)$$

where y_t is an $n \times 1$ vector of observable variables, c_t is an $n \times 1$ vector of time-varying intercepts, and $B_{i,t}$ for $i = 1, \dots, k$ are $n \times n$ matrices of time-varying coefficients. The vector of shocks u_t is heteroskedastic with variance-covariance matrix Ω_t satisfying $A_t\Omega_tA_t' = \Sigma_t\Sigma_t'$, where A_t is lower triangular and Σ_t is diagonal. All parameters of the model follow

¹⁷[Laubach and Williams \(2003\)](#) provide two estimates of the natural rate using a one-sided and a two-sided filter. [Table D.1](#) reports the results for the one-sided measure. The results are robust when the two-sided estimate is used.

random-walk processes. Further details on the estimation and specification are provided in [Appendix E](#).

We estimate the model including the variables in the following order:¹⁸ public debt-to-GDP ratio, real GDP growth, CPI inflation, and the real interest rate. [Lubik and Matthes \(2015\)](#) consider the same specification but exclude the public debt-to-GDP ratio.¹⁹ We adopt the same priors as [Primiceri \(2005\)](#), described in [Appendix E](#), and include one lag of the observable variables.²⁰

Once the model is estimated, we follow [Lubik and Matthes \(2015\)](#) and construct the natural rate of interest as the five-year-ahead forecast of the real interest rate, under the assumption that shocks continue to affect the economy but nominal frictions have disappeared.²¹ [Figure E.1](#) plots our estimate of the natural rate of interest (green line) alongside those of [Lubik and Matthes \(2015\)](#) and [Holston et al. \(2017\)](#) (blue and red lines, respectively). The three series move closely together during the early part of the sample. However, after 2010, our estimate declines more sharply, reaching much lower and even negative levels during the 2010–2020 period. This pattern is consistent with the zero lower bound episode, when nominal rates were constrained and, as shown by [Wu and Xia \(2016\)](#), the shadow policy rate turned negative. Hence, it is plausible that the natural real interest rate was also negative during this period. Supporting this interpretation, recent estimates by [Lubik and Matthes \(2023\)](#) similarly report negative values for the natural rate in the 2010s.

The debt-augmented measure of the natural rate constructed above allows us to revisit the empirical relationship between public debt and the policy rate. Our hypothesis is that public debt-to-GDP ratio appears as a relevant variable in the Taylor rule because it contains information about the natural rate of interest beyond what is captured by inflation and the

¹⁸The ordering of variables matters in any TVP–VAR because the matrix A_t is assumed to be triangular. The same issue arises in standard structural VARs identified through a Cholesky decomposition of the covariance matrix of the innovations. Reassuringly, however, our results are robust to alternative orderings.

¹⁹They also use PCE instead of CPI inflation. To maintain consistency with the empirical analysis in [Section 2](#), we use CPI inflation instead of PCE inflation. Reassuringly, results are not affected by this choice.

²⁰Adding a fourth variable increases the number of estimated parameters substantially. For this reason, our baseline specification uses one lag, although results are similar when two lags are included.

²¹The real interest rate is obtained as the difference between the nominal interest rate and inflation expectations, where the latter are computed following the methodology of [Laubach and Williams \(2003\)](#). Specifically, they proxy inflation expectations with the forecast of the four-quarter-ahead percentage change in the core PCE price index—personal consumption expenditures excluding food and energy—generated from a univariate AR(3) estimated over the prior sample.

output gap. If this is the case, controlling for our new measure of the natural rate should fully absorb the effect of public debt on the policy rate.

Table 3: Taylor rule estimates with our measure of the natural rate of interest

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
π_t	1.12*** (0.16)	1.12*** (0.09)	1.13*** (0.07)	1.07*** (0.09)	1.12*** (0.07)
y_t	0.26* (0.16)	0.09 (0.06)	0.20*** (0.07)	0.19** (0.09)	0.19*** (0.05)
d_t	-0.16 (0.22)	0.13 (0.09)	-0.02 (0.11)	0.01 (0.10)	-0.09 (0.19)
$R_t^{*,d}$	1.20*** (0.37)	1.69*** (0.13)	1.43*** (0.15)	1.50*** (0.18)	1.30*** (0.23)
ρ	0.54*** (0.17)	0.28*** (0.10)	0.42*** (0.06)	0.44*** (0.09)	0.50*** (0.03)
N	162	159	162	162	144

Notes: The table reports estimates of [equation \(2\)](#). The dependent variable is the federal funds rate, r_t . The explanatory variables are CPI inflation, π_t , the output gap, y_t , the public debt-to-GDP ratio, d_t , and our estimate of natural rate of interest, $R_t^{*,d}$. The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. The variable d_t is rescaled so that one unit corresponds to 10 percentage points. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of [Clarida et al. \(2000\)](#), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of [Angeletos et al. \(2020\)](#) and the excess bond-premium shock of [Gilchrist and Zakrajšek \(2012\)](#). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of [Ramey \(2016\)](#), and the news oil-price shocks of [Känzig \(2021\)](#). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

To test this implication, [Table 3](#) reports the estimates of [equation \(2\)](#) when our debt-informed natural rate is included as a control. The results show that the coefficient on the public debt-to-GDP ratio is no longer statistically significant, except in the last column. The coefficients on the natural rate and inflation are positive and statistically significant. Notably, the coefficient on the natural rate is close to one, implying that nominal interest rates move almost one-for-one with the natural rate. Finally, the coefficient on the output gap is positive but not always significant, consistent with the idea that it is difficult to disentangle the Federal Reserve’s response to movements in the output gap and the natural

rate, as the two contain correlated—though distinct—information.

These results provide supporting evidence that the Federal Reserve lowers the policy rate when public debt increases because debt conveys information about the natural rate, rather than because the Federal Reserve responds directly to debt. They also suggest that incorporating information on public debt is important when constructing measures of the natural rate of interest.

For this reason, and because the natural rate—rather than public debt itself—is the object of interest for the Federal Reserve, we estimate [equation \(1\)](#) using our measure of the natural rate in place of the public debt-to-GDP ratio. [Table 4](#) reports the results. The estimated coefficients on inflation and the natural rate are both positive and statistically significant. The coefficient on the natural rate is close to 1.4, indicating that a one-percent increase in the natural rate is associated with roughly a 1.4-percent increase in the nominal interest rate. The coefficient on the output gap is again positive but not always significant. Overall, these findings provide additional evidence that the Federal Reserve targets a measure of the natural rate that embeds information about public debt.

4.2.1 Robustness of the results

The results in [Table 3](#) are robust to a wide range of empirical checks. First, we construct an alternative measure of the natural rate using PCE inflation in the TVP–VAR. We then re-estimate [equation \(2\)](#) using this alternative measure as a control and replace CPI inflation with PCE inflation for consistency. [Table E.1](#) shows that results are unaffected by this choice.

Next, we estimate the TVP–VAR including the nominal interest rate directly and obtain the real rate as the difference between the forecasts of the nominal rate and inflation. [Table E.2](#) presents the results, which confirm that the coefficient on the debt-to-GDP ratio remains insignificant, while the coefficient on the natural rate is positive and statistically significant. We also re-estimate our benchmark TVP–VAR with two lags ([Table E.3](#)). Finally, the findings are robust to alternative priors for the means and variances of the random-walk processes governing the model’s parameters.²²

²²For example, we employed different priors for the variance of innovations in residual volatility and in the evolution of the VAR coefficients. Results are available upon request.

Table 4: Taylor rule with our measure of the natural rate and without public debt

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
π_t	1.17*** (0.15)	1.10*** (0.08)	1.14*** (0.05)	1.08*** (0.10)	1.13*** (0.07)
y_t	0.25* (0.15)	0.17** (0.08)	0.20*** (0.08)	0.24** (0.11)	0.18*** (0.05)
$R_t^{*,d}$	1.40*** (0.21)	1.48*** (0.10)	1.44*** (0.07)	1.45*** (0.13)	1.40*** (0.06)
ρ	0.54*** (0.17)	0.32*** (0.11)	0.44*** (0.06)	0.48*** (0.09)	0.50*** (0.03)
N	162	159	162	162	144

Notes: The table reports estimates of [equation \(1\)](#), where the debt-to-GDP ratio is replaced with our debt-informed measure of the natural rate of interest, $R_t^{*,d}$. The dependent variable is the federal funds rate, r_t . The explanatory variables are CPI inflation, π_t , the output gap, y_t , and our estimate of the natural rate of interest, $R_t^{*,d}$. The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of [Clarida et al. \(2000\)](#), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of [Angeletos et al. \(2020\)](#), and the excess bond-premium shock of [Gilchrist and Zakrajšek \(2012\)](#). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of [Ramey \(2016\)](#), and the news oil-price shocks of [Känzig \(2021\)](#). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

We perform the same set of robustness exercises for the regressions in [Table 4](#). [Table E.4](#), [Table E.5](#), and [Table E.6](#) report, respectively, the results obtained when using PCE inflation, the nominal rate, or two lags in estimating the TVP–VAR. Across all specifications, the coefficient on the debt-informed measure of the natural rate remains positive and statistically significant, with a magnitude relatively close to one.

Overall, these robustness exercises confirm that our main results are stable across alternative specifications, measures of inflation, and modeling choices.

5 Quantitative Model

So far, we have shown that the Federal Reserve lowers the policy rate when the public debt-to-GDP ratio increases. In [Section 3](#), we proposed a mechanism that rationalizes this

empirical fact: the Federal Reserve seeks to track the natural rate of interest, and public debt provides information about this unobservable variable. In [Section 4](#), we presented supporting empirical evidence consistent with this interpretation.

In [Section 3](#), we show that shocks to the demand for public debt generate a negative correlation between public debt and the natural rate, whereas tax shocks generate a positive correlation. In this section, we build and estimate a quantitative model that extends the framework of [Justiniano et al. \(2010\)](#) and incorporates the key features of the model in [Section 3](#). We use this model to show that demand shocks are the primary driver of business cycle fluctuations. We then use it to replicate our empirical findings using simulated data.

5.1 Model environment

This subsection outlines the baseline model of the U.S. business cycle. The framework is a medium-scale DSGE model with a neoclassical growth core, augmented with several standard frictions. The economy is populated by six classes of agents: producers of the final good, intermediate-goods producers, households, employment agencies, a fiscal authority, and a monetary authority. The decision problems of final-good producers, intermediate-goods producers, and employment agencies, as well as the market-clearing conditions, follow [Justiniano et al. \(2010\)](#) and are described in detail in [Appendix F.1](#). Below, we present the problems of the households, the fiscal authority, and the monetary authority.

Variables without a time script represents their steady-state levels.

5.1.1 Households problem

Each household j maximizes the expected lifetime utility

$$\mathbb{E}_t \sum_{s=0}^{\infty} \beta^s b_{t+s} \left[\log(C_{t+s} - hC_{t+s-1}) - \varphi \frac{L_{t+s}(j)^{1+\nu}}{1+\nu} + \phi_B (B_{t+s} - \eta_{t+s} B)^2 \right], \quad (4)$$

where C_t denotes consumption, h is the degree of habit formation, $L_t(j)$ is hours worked by household j , and B_t is the holding of government bonds. The parameter $\phi_B > 0$ governs the strength of households' preference for holding government bonds. Consumption and bond holdings are not indexed by j because the presence of complete state-contingent securities

ensures that, in equilibrium, all households choose identical consumption and asset positions.

The novelty in [equation \(4\)](#) relative to the standard literature is the inclusion of the term $\phi_B(B_t - \eta_t B)^2$, which captures the idea that households value government bonds not only for their pecuniary returns but also for the services they provide. This assumption has two key implications. First, it breaks Ricardian equivalence, since public debt directly affects households' decisions. Second, it establishes a direct link between the level of public debt and interest rates. This feature motivates our use of the level of public debt in the empirical analyses presented in [Section 2](#) and [Section 4](#).

Beyond these theoretical implications, the term also has a clear economic interpretation. As emphasized by [Krishnamurthy and Vissing-Jorgensen \(2012\)](#), government debt is both safe and liquid. The additional utility term can therefore be viewed as capturing the liquidity and safety services that government bonds provide beyond their financial return. We thus refer to η_t as a *liquidity* or *safety* shock. This shock evolves according to

$$\log(\eta_t) = (1 - \rho_\eta)\log(\eta) + \rho_\eta \log(\eta_{t-1}) + \varepsilon_{\eta,t}, \quad \varepsilon_{\eta,t} \stackrel{i.i.d.}{\sim} \mathcal{N}(0, \sigma_\eta^2),$$

and represents exogenous shifts in households' demand for safe and liquid assets. An increase in η_t raises the utility derived from holding government bonds, inducing households to save more and consume less, thereby generating a recessionary effect.

This shock admits several interpretations. It microfounds the *risk-premium* shock of [Smets and Wouters \(2007\)](#) when public debt supply is zero ([Fisher, 2015](#)); it corresponds to the *spread* or *flight-to-safety* shocks commonly studied in the international literature ([Eichenbaum et al., 2021](#); [Kekre and Lenel, 2024](#)); and it can be viewed as capturing increases in idiosyncratic risk ([Aiyagari, 1994](#); [Bayer et al., 2019](#)) or in the riskiness of capital returns ([Li and Merkel, 2025](#)). More broadly, it can also be interpreted as a tightening of borrowing constraints, which limits households' ability to save in safe assets—here, government debt.

In addition to the liquidity shock, households are also subject to a standard intertemporal preference shock. In [equation \(4\)](#), b_t represents a shock to the discount factor, which affects both the marginal utility of consumption and the marginal disutility of labor. This shock

follows the stochastic process

$$\log b_t = \rho_b \log b_{t-1} + \varepsilon_{b,t}, \quad \varepsilon_{b,t} \stackrel{i.i.d.}{\sim} \mathcal{N}(0, \sigma_b^2),$$

and captures temporary changes in households' willingness to substitute consumption intertemporally.

Having described households' preferences and the relevant shocks, we now turn to their budget constraint. In nominal terms, the household's flow budget constraint is

$$P_t C_t + P_t T_t + P_t I_t + B_t \leq R_{t-1} B_{t-1} + Q_t(j) + \Pi_t + W_t(j) L_t(j) + r_t^k u_t \bar{K}_{t-1} - P_t a(u_t) \bar{K}_{t-1},$$

where I_t is investment, T_t is lump-sum taxes, R_t is the gross nominal interest rate, $Q_t(j)$ denotes the net cash flow from household j 's portfolio of state-contingent securities, and Π_t is the per-capita profit accruing to households from firm ownership.

Households own the capital stock and choose the utilization rate u_t , which transforms physical capital into effective capital according to

$$K_t = u_t \bar{K}_{t-1}.$$

Effective capital is rented to firms at the rate r_t^k , while the cost of utilization is given by $a(u_t)$ per unit of physical capital. In the steady state, $u = 1$, $a(1) = 0$, and $\chi \equiv a''(1)/a'(1)$ measures the curvature of the utilization cost function. Under log-linearization, χ is the only parameter of $a(u_t)$ that affects the model's dynamics.

The accumulation of physical capital follows

$$\bar{K}_t = (1 - \delta) \bar{K}_{t-1} + \mu_t \left[1 - S \left(\frac{I_t}{I_{t-1}} \right) \right] I_t,$$

where δ is the depreciation rate and $S(\cdot)$ denotes the investment adjustment-cost function. The investment-specific shock μ_t captures exogenous variation in the efficiency with which

final goods are transformed into productive capital. This shock evolves according to

$$\log \mu_t = \rho_\mu \log \mu_{t-1} + \varepsilon_{\mu,t}, \quad \varepsilon_{\mu,t} \stackrel{i.i.d.}{\sim} \mathcal{N}(0, \sigma_\mu^2),$$

where ρ_μ measures the persistence of the shock.

As in [Erceg et al. \(2000\)](#), each period a fraction ξ_w of households cannot freely adjust their wage and instead follows the indexation rule

$$W_t(j) = W_{t-1}(j), (\pi_{t-1} e^{z_{t-1}})^{\iota_w} (\pi e^\gamma)^{1-\iota_w},$$

where z_t denotes the growth rate of the exogenous technology process, $\Delta \log A_t$:

$$\Delta \log A_t \equiv z_t = (1 - \rho_z)\gamma + \rho_z z_{t-1} + \varepsilon_{z,t}.$$

The remaining fraction $(1 - \xi_w)$ of households reoptimizes its wage $W_t(j)$ by maximizing utility subject to the budget constraint and the demand for differentiated labor $L_t(j)$, which is aggregated into total labor by *employment agencies* as described in [Appendix F.1.3](#).

5.1.2 Fiscal and monetary authority

The fiscal authority's budget constraint in real terms is

$$D_t + T_t = R_{t-1} \pi_t^{-1} D_{t-1} + G_t,$$

where D_t denotes the stock of real public debt, T_t represents lump-sum taxes, and G_t is government spending. Government purchases are assumed to be an exogenous, time-varying fraction of output, as in [Justiniano et al. \(2010\)](#):

$$G_t = \left(1 - \frac{1}{g_t}\right) Y_t,$$

where g_t follows the stochastic process

$$\log g_t = (1 - \rho_g) \log g + \rho_g \log g_{t-1} + \varepsilon_{g,t}, \quad \varepsilon_{g,t} \stackrel{i.i.d.}{\sim} \mathcal{N}(0, \sigma_g^2). \quad (5)$$

This specification implies that higher values of g_t correspond to periods of higher government expenditure relative to output.

Given the assumption of bonds in the utility function, Ricardian equivalence does not hold, since the path of public debt directly affects households' choices. Therefore, the specification of the fiscal rule governing taxes, T_t , is crucial for the model's dynamics. We assume that the fiscal authority follows the tax rule, expressed in real terms,

$$T_t - \tilde{T} = \tau_y Y_t + \tau_{yy} Y_{t-1} + \tau_d D_{t-1} + \tau_R \mathbb{E}_t \pi_{t+1}^{-1} R_t G_t + \rho_T T_{t-1} + \tau_t,$$

where $\tilde{T}$ denotes the steady-state level of tax revenues. The coefficients τ_y and τ_{yy} capture the procyclicality of fiscal policy, implying that tax revenues increase when output expands, while τ_d measures the responsiveness of taxes to the outstanding stock of public debt, governing the speed of debt stabilization. Both terms are standard in the literature. The parameter τ_R governs the sensitivity of taxes to interest-rate fluctuations and, therefore, to the cost of government spending. A positive value of τ_R implies that when interest rates rise and government spending becomes more expensive, the fiscal authority raises tax revenues to maintain fiscal discipline. When $\tau_R = 0$, as typically assumed in the literature, fiscal policy is completely inelastic to interest-rate changes. The term ρ_T allows for partial adjustment or smoothing of tax revenues over time. Finally, τ_t represents an exogenous tax disturbance, which evolves according to

$$\tau_t = \rho_\tau \tau_{t-1} + \varepsilon_{\tau,t}, \quad \varepsilon_{\tau,t} \stackrel{i.i.d.}{\sim} \mathcal{N}(0, \sigma_\tau^2).$$

This shock can be interpreted as a lump-sum fiscal stimulus or consolidation measure that temporarily alters government revenues.

Fiscal and monetary policy jointly determine the dynamics of public debt and interest rates. While the fiscal rule governs how the government adjusts taxes to stabilize debt, monetary policy sets the short-term nominal rate according to the following Taylor rule:

$$\frac{R_t}{R} = \left(\frac{R_{t-1}}{R} \right)^{\rho_R} \left[\left(\frac{\mathbb{E}_t R_{t+20}^*}{R^*} \right)^{\phi_{R^*}} \left(\frac{\pi_t}{\pi} \right)^{\phi_\pi} \left(\frac{X_t}{X_t^*} \right)^{\phi_x} \right]^{1-\rho_R} \left[\frac{X_t/X_{t-1}}{X_t^*/X_{t-1}^*} \right]^{\phi_{\Delta x}} \eta_{mp,t} \quad (6)$$

where R denotes the steady-state gross nominal interest rate and R^* the steady-state natural rate. Output is defined as $X_t = C_t + I_t + G_t$, and variables with an asterisk refer to the natural (flexible-price) economy without markup shocks. Hence, $\frac{X_t}{X_t^*}$ and $\frac{X_t/X_{t-1}}{X_t^*/X_{t-1}^*}$ represent, respectively, the output gap and its growth rate. The monetary policy shock $\eta_{mp,t}$ follows the process

$$\log \eta_{mp,t} = \rho_{mp} \log \eta_{mp,t-1} + \varepsilon_{mp,t}, \quad \varepsilon_{mp,t} \stackrel{i.i.d.}{\sim} \mathcal{N}(0, \sigma_{mp}^2).$$

The novelty in [equation \(6\)](#) relative to [Justiniano et al. \(2010\)](#) and the broader DSGE literature is the inclusion of the term $\left(\frac{\mathbb{E}_t R_{t+20}^*}{R^*}\right)^{\phi_{R^*}}$, which captures the monetary authority’s response to variations in the natural rate. This feature is central to our mechanism: as shown empirically in [Section 2](#), public debt is informative about and negatively correlated with the natural rate, R_t^* . For consistency with our empirical measure—constructed as a five-year-ahead forecast in [Section 4.2](#)—we assume that the monetary authority responds to a forecast of the natural rate rather than its contemporaneous value. This assumption also reflects a realistic feature of policy behavior:²³ the natural rate is difficult to estimate in real time, and high-frequency fluctuations in its measured value may reflect noise rather than persistent movements. By responding to $\mathbb{E}_t R_{t+20}^*$, the monetary authority smooths its reaction to short-term volatility while adjusting policy rates in line with more persistent shifts in the natural rate.

5.2 Solution and bayesian inference

Because the technology process A_t follows a unit-root process, consumption, investment, capital, real wages, output, and public debt fluctuate around a stochastic balanced-growth path. We solve the model in three steps. First, we rewrite all equations in terms of detrended variables. Second, we compute the non-stochastic steady state of the transformed system and log-linearize the model around that steady state. Third, we solve the resulting linear

²³ As pointed out by Vice Chairman of the Federal Reserve Roger W. Ferguson Jr in 2004 ([Ferguson, 2004](#)): “One way of providing that benchmark is to consider what level of the real federal funds rate, if allowed to prevail for several years, would place economic activity at its potential and keep inflation low and stable. [...] This definition makes clear that the most relevant aid in policymaking is an *intermediate-run measure*, in that there may be forces at work in the shorter run that push spending away from potential output even if the real rate were pegged at this benchmark rate.” Our assumption is that the intermediate-run horizon corresponds to five years.

rational-expectations system to obtain its state-space representation.

We estimate the model in state-space form using Bayesian methods to characterize the posterior distribution of the structural parameters. The posterior combines the likelihood function with prior distributions over these parameters. The observables used to construct the likelihood are the log differences of GDP, consumption, investment, and real wages, as well as inflation, the federal funds rate, hours worked, and the log ratio of public debt to GDP.

The model does not allow for differential long-run trends between GDP and public debt. Following [Bilbiie et al. \(2023\)](#), we therefore detrend the log ratio of public debt to GDP using a band-pass filter that extracts fluctuations with periodicities shorter than 15 years. Because this series is filtered, we include measurement error in its observation equation, whereas the remaining observables are assumed to be measured without error.²⁴

The prior and posterior estimates of the structural parameters are reported in [Table F.1](#).²⁵ Most priors are consistent with those commonly used in the literature. For parameters appearing in [Justiniano et al. \(2010\)](#), we adopt nearly identical priors. For the fiscal block, we use values close to those in [Leeper et al. \(2017\)](#) when available and otherwise assign diffuse priors. Finally, we specify a prior for the bond-preference parameter, ϕ_B , to ensure that debt has non-trivial effects on households' choices. Specifically, we center the prior on a marginal propensity to consume of approximately 0.18, in line with empirical estimates. The prior is relatively tight, as unconstrained estimation tends to push ϕ_B toward very small values. In that case, tax shocks have negligible real effects and instead behave like measurement errors, accounting for fluctuations in debt without affecting equilibrium dynamics. This outcome reflects the normalization of the liquidity shock in the bond Euler equation to enter with a unit coefficient, so that ϕ_B does not scale its propagation.

In addition, the intertemporal preference, price-markup, and wage-markup shocks are normalized to enter the consumption Euler equation, the price-inflation equation, and the wage-setting equation with unit coefficients, respectively.

²⁴[Appendix F.2](#) provides details on the data and explains how the observed variables are mapped into their model counterparts.

²⁵All parameters are estimated except the quarterly depreciation rate of capital, δ , and the steady-state ratio of government spending to GDP, $(1 - 1/g)$. Following the literature, we set $\delta = 0.025$ and $(1 - 1/g) = 0.22$, as in [Justiniano et al. \(2010\)](#).

5.3 Model results

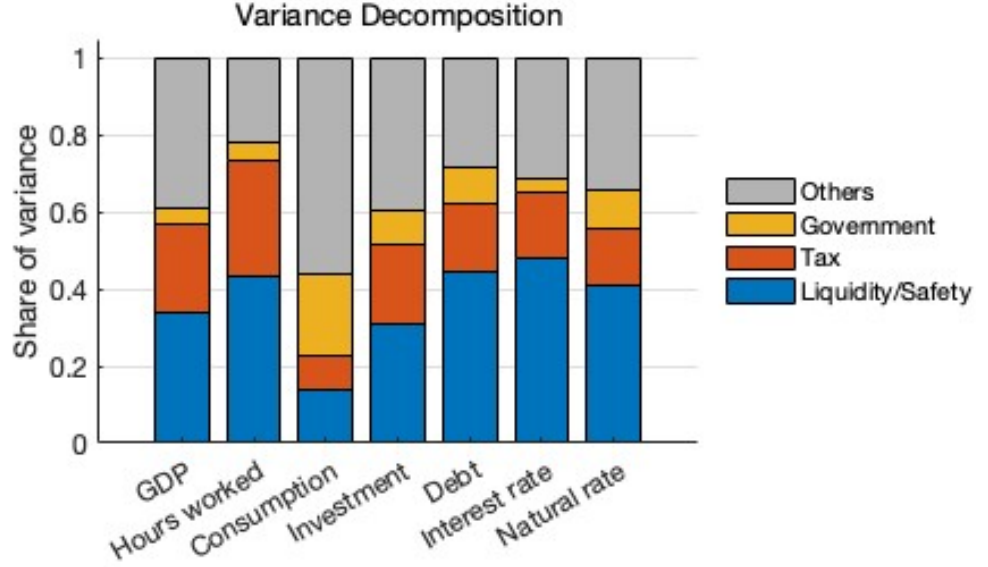
The estimated model allows us to address several key questions through a series of quantitative exercises. First, in [Section 5.3.1](#), we show that liquidity shocks account for roughly 40% of business-cycle fluctuations in GDP, public debt, the nominal interest rate, and the natural rate. Second, in [Section 5.3.2](#), we analyze the dynamic effects of a liquidity shock. Third, in [Section 5.3.3](#), we use simulated data from the model to demonstrate that our mechanism can quantitatively rationalize the empirical relationship estimated in [Section 2](#).

5.3.1 Variance decomposition

A central contribution of the paper is to show that public debt contains information about the natural rate and that the Federal Reserve responds to this information rather than to debt itself. The mechanism described in [Section 3](#) relies on shocks to households' demand for public debt, which generate a negative comovement between debt and the natural rate. When such shocks occur, an increase in debt signals a decline in the natural rate, prompting the monetary authority to lower the nominal interest rate. For this mechanism to be quantitatively relevant, liquidity shocks must account for a sizable share of the fluctuations in the nominal rate, the natural rate, and public debt.

For this reason, we compute the variance decomposition of the main macroeconomic variables implied by the estimated model, including the natural rate. [Figure 3](#) reports the contribution of the liquidity, tax and government purchase shocks to the variance of the *levels* of GDP, hours worked, consumption, investment, the public-debt-to-GDP ratio, the nominal interest rate, and the natural rate. The variance decomposition is based on the spectral representation of the model and focuses on cyclical components with periodicities between 6 and 32 quarters, following [Stock and Watson \(1999\)](#). For the first six variables—those that are observable in the data—the decomposition is computed directly from the observables. By contrast, the natural rate is unobserved, and its variance decomposition is imputed from the model's solution. To obtain the decomposition, we use the spectrum of the estimated DSGE model combined with an inverse first-difference filter for output, hours worked, consumption, and investment.

Figure 3: Posterior variance decomposition at the business cycle frequency



Notes: Business-cycle frequencies correspond to cyclical components with periods between 6 and 32 quarters. The variance decomposition is obtained from the spectral representation of the DSGE model. An inverse first-difference filter is applied to GDP, consumption, and investment to recover their level components. The spectral density is computed from the model's state-space representation using 500 frequency bins covering the range of business-cycle periodicities. The figure shows that liquidity (safety) shocks account for an important share of business-cycle fluctuations.

Figure 3 shows that shocks to households' demand for public debt (in blue) account for roughly 40% of the variation in GDP, hours worked, investment, public debt, the nominal interest rate, and the natural rate, and about 20% of the variation in consumption. These shocks therefore play a central role in explaining business-cycle fluctuations. This result is consistent with the interpretation of these shocks as the Smets and Wouters (2007) "risk-premium" shock, which also explains a large share of cyclical variation in standard DSGE models. The main novelty of our framework is that these shocks jointly drive movements in the debt-to-GDP ratio and the natural rate, creating a systematic link between the two variables. This connection allows the Federal Reserve to exploit information contained in public debt when setting the policy rate.

Figure 3 also shows that tax shocks (in red) account for part of the variation in GDP, hours worked, investment, and public debt. However, compared with liquidity shocks, they explain a smaller share of the fluctuations in the natural rate and the nominal interest rate. In canonical DSGE models such as Christiano et al. (2005); Smets and Wouters (2007);

Justiniano et al. (2010), these shocks have no effect on business-cycle dynamics because Ricardian equivalence holds. In contrast, in our framework—where Ricardian equivalence is broken—tax shocks contribute meaningfully to macroeconomic fluctuations. Government purchase shocks (in yellow) explain a smaller fraction of the variance than typically found in the DSGE literature, largely because our model also allows for tax shocks. Nevertheless, these shocks still account for a nontrivial share of the variation in consumption, around 20%.

Finally, Table F.2 reports the variance decomposition for all observable variables and for the natural rate. Technology shocks account for roughly 25% of the variation in GDP and consumption, while investment-specific shocks explain about 20% of the variability in investment. Price and wage markup shocks together explain around 70% of the variation in price and wage inflation—a larger share than typically found in the literature. However, this result is consistent with recent evidence, such as Stock and Watson (2020) and Hazell et al. (2022), documenting a flattening of the Phillips curve beginning in the 1980s, which coincides with the start of our sample period. Our estimation sample also differs from much of the DSGE literature, as it begins later and extends through 2019, potentially amplifying the relative importance of markup shocks in explaining inflation dynamics.

5.3.2 Impulse response function to a liquidity shock

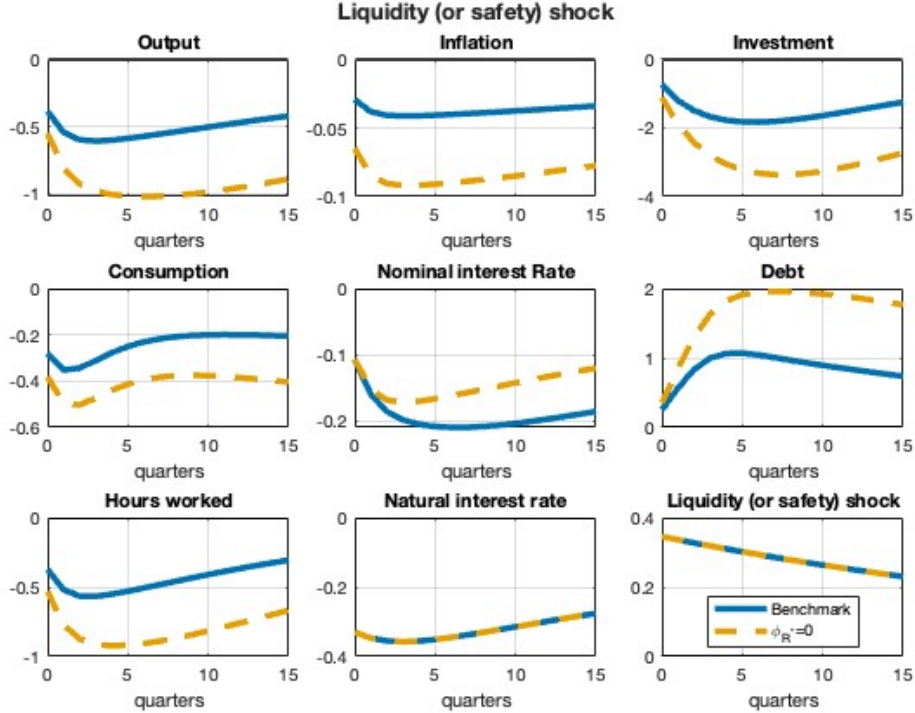
In Section 3, we showed that a positive shock to households’ demand for public debt leads to an increase in public debt and a decline in both the nominal and natural interest rates. This result provided the intuition behind our main empirical finding in Section 2: the Federal Reserve does not react to public debt per se, but to the information debt conveys about the natural rate. We now ask whether this mechanism remains valid in the estimated DSGE model that captures the main features of the U.S. economy.

For this reason, Figure 4 reports the impulse response functions (IRFs) to a liquidity shock under two alternative Taylor rules: the estimated rule (in blue) and a counterfactual rule in which the coefficient on the natural rate is set to zero (in yellow). The panels display, in order, the responses of output, inflation, investment, consumption, the nominal interest rate, public debt, hours worked, the natural rate, and the liquidity shock.

Figure 4 corroborates the intuition developed in Section 3. When a positive liquidity

shock hits the economy, the nominal interest rate falls and public debt rises. Households shift resources from consumption toward the safe asset—public debt—leading to a decline in output, investment, and hours worked. To counteract this recessionary effect, the monetary authority lowers the policy rate, regardless of whether it explicitly targets the natural rate. However, when the policy rule includes the natural rate, the nominal rate declines more sharply, providing a stronger stabilization of the economy. In this case, the contractions in output, inflation, investment, consumption, and hours worked, as well as the increase in public debt, are all more moderate. Thus, it is optimal for the monetary authority to target the natural rate in response to liquidity shocks. The same insight derived from the simple model in [Section 3](#) therefore holds in the full quantitative model, reinforcing the interpretation of our empirical findings in [Section 2](#).

Figure 4: Impulse response functions to a liquidity (or safety) shock



Notes: The figure shows the impulse response functions to a one-standard-deviation liquidity (safety) shock in the estimated DSGE model. The blue line corresponds to the estimated Taylor rule, while the yellow line corresponds to the rule with no response to the natural rate, $\phi_{R^*} = 0$. The panels display, in order, output, inflation, investment, consumption, the nominal interest rate, public debt, hours worked, marginal cost, and the natural rate of interest. The figure highlights the negative comovement between public debt and the natural rate implied by the model's liquidity shock.

The mechanism that links public debt to the natural rate also helps to illustrate how our model departs from standard DSGE frameworks. As already discussed, the liquidity shocks in our model microfound the [Smets and Wouters \(2007\)](#) “risk-premium” shock when public debt is held fixed. In that case, the mechanism linking debt and the natural rate disappears: if public debt cannot adjust, the negative correlation between the two variables vanishes, and debt no longer contains information about the natural rate. This comparison highlights that allowing debt to move endogenously is essential for our mechanism to operate.

Beyond validating the mechanism discussed above, the impulse responses also reproduce the typical dynamics of a recession. A positive liquidity shock generates a contraction in output, investment, and consumption, together with deflation and a decline in the nominal interest rate, while public debt rises. These patterns explain why liquidity shocks account for a large share of business-cycle fluctuations, as shown in [Section 5.3.1](#).

In contrast to liquidity shocks, tax shocks—reported in [Figure F.1](#)—produce similar recessionary dynamics but with opposite implications for public debt. A tax increase leads to a contraction in output, investment, and consumption, accompanied by deflation and a decline in both the natural and nominal interest rates. However, public debt falls, thereby generating a positive comovement between the natural rate and debt. When the monetary authority targets the natural rate, the declines in output, consumption, investment, and inflation are more moderate. At the same time, the reduction in public debt is more pronounced, whereas for a liquidity shock, targeting the natural rate leads to a smaller increase in debt. This asymmetry arises because debt moves together with output: lower output reduces tax revenues, leading to a smaller fiscal surplus and, consequently, a sharper fall in debt. These differences explain the distinct patterns of debt dynamics under the two Taylor rules across the two types of shocks.

5.3.3 Estimate of the Taylor rule with simulated data

We interpret the empirical result in [Section 2](#)—that the Federal Reserve lowers the nominal interest rate when the public debt-to-GDP ratio increases—not as evidence that the Federal Reserve reacts to debt itself, but as reflecting that debt provides information about movements in the natural rate, to which policy responds. In this section, we use the es-

timated DSGE model to verify whether this mechanism can quantitatively reproduce the same relationship observed in the data.

To this end, we test whether the model can replicate the empirical finding that the policy rate falls when public debt rises, even though the central bank does not react to debt directly. To do so, we simulate 2,000 artificial economies using parameters fixed at the posterior mode, each for 250 periods, discarding the first 100 observations. In these simulations, the monetary authority sets the nominal interest rate in response to the natural rate but not to public debt. We then estimate, by ordinary least squares, [equation \(6\)](#), but replacing the natural rate with the debt-to-GDP ratio. If our mechanism is correct—namely, that public debt is informative about the natural rate—the regression on simulated data should yield a negative coefficient on debt.

[Table 5](#) reports the results of this simulation exercise under several scenarios. The first column lists the true parameters used in the model. The second column presents the regression results when all shocks are used in generating the simulated data. The third column removes monetary policy shocks to address the endogeneity between the interest rate and public debt. The fourth column reports results when the economy is simulated only under liquidity shocks, and the last column when it is simulated only under fiscal shocks—that is, government spending and tax shocks.

The table reveals a clear pattern: whenever liquidity shocks are present (columns 2, 3, and 4), the coefficient on public debt is negative. In column 2, where all shocks are included, the coefficient is not statistically significant. However, this specification suffers of the endogeneity problem introduced by monetary policy shocks. Once these shocks are excluded (column 3), the estimated coefficient becomes negative and significant. When only liquidity shocks are included (column 4), the coefficient remains negative, larger in magnitude, and statistically significant. Its value, around -0.3, is less than half of the empirical estimate of -0.7 reported in [Table 1](#). Although the simulated coefficient is smaller, the fact that the model reproduces the correct sign supports our interpretation of the empirical finding. Indeed, we are estimating a misspecified regression that omits the natural rate, so a negative coefficient is not mechanically guaranteed.

On the other hand, when only fiscal shocks are included (column 5), the coefficient on

Table 5: Estimated Taylor rule using simulated data

	(1) Model	(2) All shocks	(3) No monetary shock	(4) Only liquidity shock	(5) Only fiscal shocks
R_{t-1}	0.81	0.86 (0.04)	0.88 (0.02)	0.76 (0.00)	0.68 (0.02)
y_t	0.01	0.14 (0.08)	0.13 (0.03)	-0.00 (0.01)	0.14 (0.00)
π_t	1.41	2.27 (0.75)	2.51 (0.41)	4.52 (0.08)	1.00 (0.09)
d_t	—	-0.23 (0.31)	-0.38 (0.11)	-0.27 (0.01)	0.17 (0.02)
Δy_t	0.16	0.13 (0.03)	0.22 (0.01)	0.18 (0.00)	0.16 (0.01)

Notes: The table reports ordinary least squares estimates of [equation \(6\)](#) with the natural-rate term replaced by the debt-to-GDP ratio using simulated data with parameters fixed at the posterior mode. We simulate 2,000 economies for 250 periods and discard the first 100 observations; in the simulations, policy reacts to the natural rate but not to debt. The first column shows the true parameter values implied by the model. The second column reports estimates when all structural shocks are included. The third column excludes monetary policy shocks. The fourth and fifth columns display results when the economy is simulated only under liquidity shocks and only under fiscal shocks (government spending and tax shocks), respectively. Standard errors are reported in parentheses.

public debt becomes positive. This result further supports the view that the negative relationship found in the data reflects the predominance of liquidity shocks during the analyzed period, which were the main drivers of U.S. business-cycle fluctuations, as documented in [Section 5.3.1](#).

Finally, across all specifications, the estimated coefficients on the lagged interest rate and on the growth rate of the output gap are close to the true values reported in the first column. In contrast, the coefficient on the output gap itself deviates substantially from the model-implied value. This likely reflects the close relationship between the natural rate and the output gap in the model, and the exclusion of the natural rate from the regression. As a result, the true coefficient on the output gap is difficult to recover, which helps explain why this term is often insignificant in [Section 4](#).

Together, these simulation results confirm that the estimated model can quantitatively reproduce the key empirical relationship between public debt and monetary policy documented in the data.

6 Conclusions

We document a new empirical fact: conditional on inflation and output, the Federal Reserve lowers the policy rate when the U.S. public debt-to-GDP ratio increases. We propose the following explanation. Public debt conveys information about the natural rate, and the Federal Reserve responds to this information rather than to debt itself. To formalize this mechanism, we develop a model in which debt and the natural rate are jointly determined. In the model, shocks to households’ demand for public debt generate a negative comovement between debt and the natural rate, consistent with the evidence.

We corroborate this mechanism both empirically and quantitatively. Empirically, we construct a debt-informed measure of the natural rate and show that once this measure is included in the Taylor rule, the estimated response of monetary policy to debt disappears. Monetary policy responds to the natural rate, not to debt. Quantitatively, our estimated DSGE model shows that shocks to households’ demand for public debt account for a large share of business-cycle fluctuations and reproduce the observed negative comovement between debt and the natural rate.

Although the mechanism accounts for the typical negative relationship between public debt and the natural rate, it also clarifies when this relationship reverses. If debt rises because of expansionary fiscal policy, through higher government spending or lower taxes, the natural rate increases, prompting the Federal Reserve to raise the policy rate rather than lower it. The negative comovement between debt and the policy rate therefore reflects the nature of the underlying shocks, not systematic fiscal accommodation.

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Online Appendix

Why the Federal Reserve Cuts Rates when Public Debt Rises

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A Data

In this section, we provide details on the variables used. [Table A1](#) lists all the variables used, their description and the source.

Table A1: Data and Sources

Variable	Description	Source
Federal Funds rate	Fed’s policy rate (code: FEDFUNDS)	FRED
Shadow rate	Fed’s policy rate for the ZLB period from Wu and Xia (2016)	Wu and Xia (2016)
Consumer Price Index	Price index (code: CPILFESL)	FRED
Output gap	Difference between GDP and potential GDP from CBO	CBO
Public Debt	Market value of marketable Treasury debt (code: MVGFD027MNFRBDAL)	FRED
GDP	Gross domestic product (code: GDP)	FRED
Commodity price inflation	Commodity prices are a composite of commodity goods (including oil, gold and food items)	Carvalho, Nechio, and Tristao (2021)
Money growth	Percentage change in M2 (code: M2SL)	FRED
Spread	Spread between the 10-year and the 3-month Treasury bill rates (code: TB3MS & IRLTLT01USM156N)	FRED
“Main business cycle” shock	Shock identified in a VAR using a max-share approach on unemployment	Ferrara (2026) ²⁶
“Oil supply news” shock	Shock identified with proxy-SVAR using high-frequency surprises as instrument	Känzig (2021)
“Excess bond premium” shock	Shock identified in a VAR with recursive ordering	Gilchrist and Zakrajšek (2012)

Continued on next page

²⁶[Ferrara \(2026\)](#) extended the sample period of [Angeletos, Collard, and Dellas \(2020\)](#).

Table A1 – Continued

Variable	Description	Source
“Government spending” shock	Shock identified with an internal VAR, ordering the instrument first	Ramey (2016)
Personal Consumption Expenditure	Price index (code: JCXFE)	FRED
GDP Deflator	Price index (code: A191RI1Q225SBEA)	FRED
Greenbook forecasts	Internal Fed’s forecast on inflation and output	Philadelphia FED
# Pres-Fed interactions	Number of President-Fed calls	Drechsel (2024)
Distance to elections	Quarters to the next midterm or presidential elections	Own calculations
NBER recession dates	Equal to 1 if in a recession; 0 otherwise (code: USRECD)	FRED
BAA-AAA spread	Difference between BAA and AAA corporate bond rates (codes: DBAA and DAAA)	FRED
GZ spread	Measure of spread between corporate bond yield and Treasury yield	Gilchrist and Zakrajšek (2012)
NFCI Index	Financial condition index	Chicago Fed
VIX	Market expectation of near term volatility conveyed by stock index option prices (code: VIXCLS)	FRED
S&P 500	Standard and Poor’s Stock Market Index	Bloomberg
$R_t^{*,lw}$	Model-based estimate of the natural rate	Laubach and Williams (2003)
$R_t^{*,lm}$	Estimate of natural rate from time-varying VAR	Lubik and Matthes (2015)
$R_t^{*,hlw}$	Model-based estimate of the natural rate	Holston, Laubach, and Williams (2017)

Continued on next page

Table A1 – Continued

Variable	Description	Source
$R_t^{*,d}$	Estimate of the natural rate from time-varying VAR including the debt-to-GDP ratio	Own estimation

B Empirical Robustness Checks

In this section, we present several robustness checks for [Table 1](#). First, we show that the empirical finding is robust to alternative variable definitions and sample choices. Second, we examine whether the result is driven by low-frequency movements in trending variables. Third, we evaluate several potential mechanisms that could rationalize the finding and show that the data do not support them.

In estimating [equation \(1\)](#), we use CPI inflation, the CBO output gap, and the market value of gross federal debt relative to GDP. As a measure of the policy rate, we use the federal funds rate, replacing it with the shadow rate during the zero lower bound period ([Wu and Xia, 2016](#)).²⁷

In [Table 1](#), we employ several estimation techniques. Column (1) reports OLS estimates. Column (2) presents Generalized Method of Moments (GMM) estimates using four lags of the independent variables, commodity price inflation, money growth, and the spread between the long-term bond rate and the three-month Treasury bill as instruments.²⁸ Column (3) uses GMM with four lags of the independent variables and lags five through twenty of the business-cycle shock of [Angeletos, Collard, and Dellas \(2020\)](#) as additional instruments, while Column (4) replaces this shock with the excess bond-premium shock of [Gilchrist and Zakrajšek \(2012\)](#). Finally, Column (5) uses only lagged structural shocks as instruments: the business-cycle shock, the excess bond-premium shock, the oil news shocks of [Känzig \(2021\)](#), and the fiscal shocks of [Ramey \(2016\)](#).

Variable definitions and samples. We examine whether the results are sensitive to alternative variable definitions. [Table B1](#) and [Table B2](#) show that using different measures of inflation,

²⁷We define the zero lower bound period as 2009:Q1–2015:Q4.

²⁸These instruments follow [Clarida, Gali, and Gertler \(2000\)](#).

respectively GDP deflator or PCE inflation, does not alter the findings.²⁹ The same conclusion holds when we use the federal funds rate without replacing it with the shadow rate of Wu and Xia (2016) (Table B3). To assess the extent to which the results are driven by the zero lower bound period, we also stop the sample in 2011:Q3. As shown in Table B4, the results remain qualitatively similar, although the estimated coefficient on the public debt-to-GDP ratio is larger in magnitude (approximately 1.2).

Because the regression includes the debt-to-GDP ratio, one might worry that the results are driven by movements in the denominator rather than the numerator. To address this concern, we replicate the analysis by scaling debt by lagged GDP (at time $t - 1$) or by potential output at time t . Table B5 and Table B6 show that the results are unaffected by these alternative normalizations.

Finally, equation (1) specifies the Federal Reserve’s reaction function as reacting to variables at time t . However, as emphasized by Clarida, Gali, and Gertler (2000), central banks behave in a forward-looking manner, responding to expected future inflation and output. Accordingly, in Table B7 we re-estimate equation (1) using realized inflation and output gap at time $t + 1$. The results remain robust under this alternative specification.

²⁹Using PCE inflation yields a noisier estimate of the coefficient on the public debt-to-GDP ratio, but the results remain broadly consistent with those reported in Table 1.

Table B1: Taylor rule estimates using the GDP deflator instead of the CPI

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
π_t^{gdp}	1.29*** (0.39)	1.85*** (0.57)	1.93*** (0.40)	1.60*** (0.40)	1.42*** (0.26)
y_t	0.71*** (0.24)	0.81*** (0.24)	1.00*** (0.23)	0.86*** (0.21)	0.58*** (0.15)
d_t	-0.82*** (0.27)	-0.69** (0.27)	-0.40* (0.24)	-0.58*** (0.23)	-1.08*** (0.14)
ρ	0.76*** (0.09)	0.86*** (0.04)	0.84*** (0.03)	0.86*** (0.04)	0.78*** (0.04)
N	162	159	162	162	144

Notes: The table reports estimates of [equation \(1\)](#). The dependent variable is the federal funds rate, r_t . The explanatory variables are inflation, constructed using the GDP deflator, π_t , the output gap, y_t , and the public debt-to-GDP ratio, d_t . The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. The variable d_t is rescaled so that one unit equals 10 percentage points. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of [Clarida, Gali, and Gertler \(2000\)](#), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of [Angeletos, Collard, and Dellas \(2020\)](#) and the excess bond-premium shock of [Gilchrist and Zakrajšek \(2012\)](#). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of [Ramey \(2016\)](#), and the news oil-price shocks of [Känzig \(2021\)](#). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B2: Taylor rule estimates using PCE inflation instead of CPI

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
π_t^{pce}	1.60*** (0.33)	2.17*** (0.28)	2.10*** (0.24)	2.01*** (0.28)	1.69*** (0.20)
y_t	0.78*** (0.19)	1.12*** (0.18)	1.06*** (0.15)	1.16*** (0.18)	0.70*** (0.12)
d_t	-0.59*** (0.23)	-0.25 (0.17)	-0.26* (0.15)	-0.21 (0.17)	-0.76*** (0.12)
ρ	0.68*** (0.10)	0.77*** (0.04)	0.74*** (0.03)	0.79*** (0.04)	0.70*** (0.04)
N	162	159	162	162	144

Notes: The table reports estimates of [equation \(1\)](#). The dependent variable is the federal funds rate, r_t . The explanatory variables are PCE inflation, π_t , the output gap, y_t , and the public debt-to-GDP ratio, d_t . The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. The variable d_t is rescaled so that one unit equals 10 percentage points. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of [Clarida, Gali, and Gertler \(2000\)](#), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of [Angeletos, Collard, and Dellas \(2020\)](#) and the excess bond-premium shock of [Gilchrist and Zakrajšek \(2012\)](#). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of [Ramey \(2016\)](#), and the news oil-price shocks of [Känzig \(2021\)](#). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B3: Taylor rule estimates without using the Wu and Xia (2016) shadow rate

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
π_t^{cpi}	1.17*** (0.24)	1.20*** (0.20)	1.25*** (0.14)	1.08*** (0.20)	1.30*** (0.10)
y_t	0.47*** (0.17)	0.55*** (0.14)	0.59*** (0.11)	0.62*** (0.13)	0.38*** (0.10)
d_t	-0.62*** (0.18)	-0.54*** (0.13)	-0.48*** (0.11)	-0.54*** (0.13)	-0.54*** (0.08)
ρ	0.66*** (0.11)	0.75*** (0.05)	0.68*** (0.03)	0.76*** (0.05)	0.66*** (0.03)
N	162	159	162	162	144

Notes: The table reports estimates of equation (1). The dependent variable is the federal funds rate, r_t , and, unlike in the baseline specification, it is not replaced by the Wu and Xia (2016) shadow rate for the period from 2009:Q1 to 2015:Q4. The explanatory variables are CPI inflation, π_t , the output gap, y_t , and the public debt-to-GDP ratio, d_t . The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. The variable d_t is rescaled so that one unit equals 10 percentage points. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of Clarida, Gali, and Gertler (2000), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of Angeletos, Collard, and Dellas (2020) and the excess bond-premium shock of Gilchrist and Zakrajšek (2012). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of Ramey (2016), and the news oil-price shocks of Känzig (2021). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B4: Taylor rule estimates for the pre-zero lower bound period

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
π_t^{cpi}	0.85*** (0.23)	0.83*** (0.20)	1.01*** (0.16)	0.72*** (0.19)	1.06*** (0.11)
y_t	0.43*** (0.16)	0.50*** (0.14)	0.51*** (0.12)	0.55*** (0.13)	0.42*** (0.09)
d_t	-1.35*** (0.26)	-1.24*** (0.18)	-1.06*** (0.20)	-1.36*** (0.24)	-1.35*** (0.12)
ρ	0.58*** (0.14)	0.70*** (0.06)	0.61*** (0.05)	0.69*** (0.06)	0.56*** (0.05)
N	130	130	130	130	128

Notes: The table reports estimates of [equation \(1\)](#) over a shortened sample ending in 2011:Q3, when the zero lower bound became binding. The dependent variable is the federal funds rate, r_t . The explanatory variables are CPI inflation, π_t , the output gap, y_t , and the public debt-to-GDP ratio, d_t . The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. The variable d_t is rescaled so that one unit equals 10 percentage points. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of [Clarida, Gali, and Gertler \(2000\)](#), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of [Angeletos, Collard, and Dellas \(2020\)](#) and the excess bond-premium shock of [Gilchrist and Zakrajšek \(2012\)](#). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of [Ramey \(2016\)](#), and the news oil-price shocks of [Känzig \(2021\)](#). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B5: Taylor rule estimates using the ratio of debt at time t to GDP at time $t - 1$

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
π_t^{cpi}	1.19*** (0.25)	1.11*** (0.23)	1.30*** (0.17)	1.20*** (0.22)	1.19*** (0.12)
y_t	0.61*** (0.20)	0.73*** (0.17)	0.84*** (0.16)	0.82*** (0.16)	0.46*** (0.11)
d_t	-0.72*** (0.22)	-0.80*** (0.18)	-0.55*** (0.16)	-0.60*** (0.17)	-0.93*** (0.10)
ρ	0.68*** (0.10)	0.76*** (0.05)	0.72*** (0.03)	0.77*** (0.04)	0.63*** (0.04)
N	162	159	162	162	144

Notes: The table reports estimates of [equation \(1\)](#). The dependent variable is the federal funds rate, r_t . The explanatory variables are CPI inflation, π_t , the output gap, y_t , and ratio of public debt at time t to GDP at time $t - 1$, d_t . The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. The variable d_t is rescaled so that one unit equals 10 percentage points. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of [Clarida, Gali, and Gertler \(2000\)](#), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of [Angeletos, Collard, and Dellas \(2020\)](#) and the excess bond-premium shock of [Gilchrist and Zakrajšek \(2012\)](#). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of [Ramey \(2016\)](#), and the news oil-price shocks of [Känzig \(2021\)](#). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B6: Taylor rule estimates using the ratio of debt to potential GDP

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
π_t^{cpi}	1.21*** (0.25)	1.17*** (0.23)	1.34*** (0.17)	1.26*** (0.22)	1.21*** (0.12)
y_t	0.67*** (0.19)	0.80*** (0.17)	0.91*** (0.16)	0.89*** (0.16)	0.52*** (0.10)
d_t	-0.71*** (0.22)	-0.76*** (0.18)	-0.52*** (0.16)	-0.54*** (0.17)	-0.94*** (0.10)
ρ	0.69*** (0.10)	0.77*** (0.05)	0.73*** (0.03)	0.78*** (0.04)	0.65*** (0.04)
N	162	159	162	162	144

Notes: The table reports estimates of [equation \(1\)](#). The dependent variable is the federal funds rate, r_t . The explanatory variables are CPI inflation, π_t , the output gap, y_t , and the ratio of public debt to potential GDP, d_t . The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. The variable d_t is rescaled so that one unit equals 10 percentage points. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of [Clarida, Gali, and Gertler \(2000\)](#), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of [Angeletos, Collard, and Dellas \(2020\)](#) and the excess bond-premium shock of [Gilchrist and Zakrajšek \(2012\)](#). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of [Ramey \(2016\)](#), and the news oil-price shocks of [Känzig \(2021\)](#). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B7: Taylor rule estimates using inflation and output gap at time $t + 1$

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
π_{t+1}^{cpi}	1.51*** (0.30)	1.46*** (0.18)	1.63*** (0.15)	1.40*** (0.21)	1.41*** (0.10)
y_{t+1}	0.63*** (0.19)	0.81*** (0.15)	0.79*** (0.13)	0.90*** (0.16)	0.40*** (0.08)
d_t	-0.59*** (0.20)	-0.58*** (0.15)	-0.44*** (0.13)	-0.56*** (0.14)	-0.96*** (0.07)
ρ	0.70*** (0.08)	0.74*** (0.04)	0.73*** (0.03)	0.78*** (0.04)	0.60*** (0.02)
N	162	159	162	162	144

Notes: The table reports estimates of [equation \(1\)](#). The dependent variable is the federal funds rate, r_t . The explanatory variables are CPI inflation, π_{t+1} , the output gap, y_{t+1} , and the public debt-to-GDP ratio, d_t . The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. The variable d_t is rescaled so that one unit equals 10 percentage points. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of [Clarida, Gali, and Gertler \(2000\)](#), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of [Angeletos, Collard, and Dellas \(2020\)](#) and the excess bond-premium shock of [Gilchrist and Zakrajšek \(2012\)](#). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of [Ramey \(2016\)](#), and the news oil-price shocks of [Känzig \(2021\)](#). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Controlling for low-frequency variation. We run a series of robustness checks to make sure our result is not driven by low-frequency variation in trending variables. [Table B8](#) shows the [Cumby and Huizinga \(1992\)](#) serial correlation test for each column of [Table 1](#). Reassuringly, we cannot reject the null hypothesis that residuals are not correlated up to four lags.

Table B8: Cumby and Huizinga (1992) serial correlation test

	OLS	GMM	GMM BC	GMM EBP	GMM Shocks
1	.663	.158	.768	.309	.490
2	.197	.281	.601	.494	.607
3	.095	.253	.323	.306	.565
4	.168	.384	.444	.459	.646

Notes: Each row j reports the p-value of Cumby and Huizinga (1992) test statistic for serial correlation in the residuals under the null that all autocorrelation coefficients up to j are zero. Each column corresponds to the respective specification in Table 1.

We next estimate equation (1) using filtered variables. The results are reported in Table B9. As in Table 1, all coefficients are scaled by $1 - \rho_1 - \rho_2 \equiv 1 - \rho$, so that they can be interpreted as long-run effects. Column (1) reports the OLS estimates without detrending for comparison. Columns (2), (4) and (6) present OLS estimates under alternative detrending methods, while Columns (3), (5) and (7) use lagged macroeconomic shocks—those employed in the final column of Table 1—as instruments. We rely on these shocks as instruments because they exhibit no trend and therefore do not require detrending. Columns (2) and (3) apply the one-sided Hodrick and Prescott (1997) filter. Columns (4) and (5) use the Hamilton (2018) filter. Columns (6) and (7) use a band-pass filter that extracts fluctuations with periodicities shorter than 15 years.

The coefficient on debt is statistically significant, with a magnitude of approximately 0.25. For comparison, the coefficient on inflation is around 0.45, indicating that debt has an economically meaningful influence on Federal Reserve policy decisions. The coefficient on the output gap is also positive, as expected, though not always statistically significant, with a magnitude of about 0.3—very close to that of debt.

Table B9: Taylor rule estimates using filtered variables

	(1) Benchmark	(2) OLS HP	(3) GMM HP	(4) OLS Ham	(5) GMM Ham	(6) OLS BP	(7) GMM BP
π_t^{cpi}	0.65*** (0.14)	0.51*** (0.14)	0.42*** (0.06)	0.63*** (0.15)	0.49*** (0.05)	0.46*** (0.17)	0.40*** (0.07)
y_t	0.27*** (0.09)	0.31 (0.19)	0.34*** (0.07)	0.28 (0.20)	0.05 (0.06)	0.42*** (0.16)	0.46*** (0.10)
d_t	-0.41*** (0.12)	-0.35** (0.14)	-0.35*** (0.05)	-0.32* (0.18)	-0.44*** (0.06)	-0.35*** (0.11)	-0.39*** (0.09)
ρ	0.69*** (0.11)	0.49*** (0.16)	0.37*** (0.05)	0.62*** (0.10)	0.34*** (0.05)	0.52*** (0.14)	0.40*** (0.05)
N	162	162	144	162	144	162	144

Notes: The table reports estimates of [equation \(1\)](#). The dependent variable is the federal funds rate, r_t . The explanatory variables are CPI inflation, π_{t+1} , the output gap, y_{t+1} , and the public debt-to-GDP ratio, d_t . The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. All variables are standardized by their respective standard deviations. Column (1) reports OLS estimates when the variables are in levels. Columns (2) and (3) report OLS and GMM estimates when all variables are detrended using the one-sided [Hodrick and Prescott \(1997\)](#) filter. Columns (4) and (5) report OLS and GMM estimates when all variables are detrended using the [Hamilton \(2018\)](#) filter. Columns (6) and (7) report OLS and GMM estimates when all variables are detrended using a band-pass filter that extracts fluctuations with periodicities shorter than 15 years. Columns (3), (5) and (7) use as instruments lags 5 through 20 of the business-cycle shock of [Angeletos, Collard, and Dellas \(2020\)](#), the excess bond premium shock of [Gilchrist and Zakrajšek \(2012\)](#), the government spending shock of [Ramey \(2016\)](#), and the news oil-price shocks of [Känzig \(2021\)](#). Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

As an additional exercise, in [Table B10](#), we employ the system-projections instrumental-variable (SP-IV) method of [Lewis and Mertens \(2022\)](#), which estimates the Taylor rule in the impulse-response-function space, where variables do not exhibit trends. We again use the same macroeconomic shocks as in column (5) of [Table 1](#) as instruments. [Table B10](#) shows that the coefficient on the debt-to-GDP ratio remains negative and statistically significant.

Table B10: Taylor rule estimates using the system-projections instrumental-variables method

	(1)	(2)	(3)	(4)
π_t^{cpi}	0.65 (0.36)	0.72 (0.34)	0.43 (0.21)	0.59 (0.21)
y_t	0.73 (0.18)	0.61 (0.25)	0.68 (0.13)	0.72 (0.17)
d_t	-1.02 (0.23)	-1.29 (0.32)	-1.01 (0.22)	-0.66 (0.20)
ρ	0.65 (0.08)	0.68 (0.08)	0.41 (0.10)	0.49 (0.10)
two lags of instrument	No	Yes	No	Yes
two lags of regressors	No	No	Yes	Yes

Notes: The table reports estimates of [equation \(1\)](#) using the system-projections instrumental-variables method of [Lewis and Mertens \(2022\)](#). The dependent variable is the federal funds rate, r_t . The explanatory variables are PCE inflation, π_t , the output gap, y_t , the public debt-to-GDP ratio, d_t . The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. The variable d_t is rescaled so that one unit corresponds to 10 percentage points. The instruments are the shocks of [Angeletos, Collard, and Dellas \(2020\)](#), [Gilchrist and Zakrajsek \(2012\)](#), [Ramey \(2016\)](#), and [Känzig \(2021\)](#). The first column includes no controls. The second column includes two lags of the instruments as controls. The third column includes two lags of the regressors as controls. The fourth column includes two lags of both the instruments and the regressors as controls.

Ruling out alternative explanations. In this section, we show that our finding cannot be explained by a range of alternative hypotheses. We augment [equation \(1\)](#) by introducing, one at a time, additional regressors that the debt-to-GDP ratio may proxy for, and show that the estimated effect remains robust.

First, public debt may proxy for near-term expected inflation and output. [Table B11](#) replaces contemporaneous variables with the average of inflation and the output gap over the subsequent four quarters. We replicate this exercise using Greenbook forecasts over the same horizon ([Table B12](#)).

Second, public debt may contain information about inflation and the output gap at longer horizons. [Table B13](#), [Table B14](#), and [Table B15](#) report results from specifications that include three-, five-, and ten-year-ahead realized values of inflation and the output gap as additional controls.

Third, public debt may enter significantly because of political pressure on the Federal Reserve to accommodate fiscal conditions. To examine this possibility, we add a measure of U.S. President–Fed interactions from [Drechsel \(2024\)](#) ([Table B16](#)), as well as a variable capturing the number of quarters

until the next midterm or presidential election (Table B17).

Fourth, public debt may more broadly reflect cyclical economic conditions. During recessions, fiscal expansions mechanically increase deficits and debt, while policy rates typically decline. This mechanism could therefore generate the negative relationship we document. We test this hypothesis in Table B18.

Finally, public debt may proxy for financial conditions. There is evidence that the Federal Reserve responds to financial market developments (Caballero, Caravello, and Simsek, 2024). We therefore control for the BAA–AAA spread, the Gilchrist and Zakrajšek (2012) spread, and the Chicago Fed National Financial Conditions Index (Table B19, Table B20, Table B21). We also include stock market indicators, such as the CBOE Volatility Index and S&P 500 returns (Table B22, Table B23).

Across all specifications, the coefficient on the public debt-to-GDP ratio remains negative and statistically significant, suggesting that a different channel is at play.

Table B11: Taylor rule estimates using average realized values of inflation and output

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
$\sum_{j=0}^4 \pi_{t+j}$	1.67*** (0.48)	1.64*** (0.21)	1.91*** (0.22)	1.66*** (0.31)	1.49*** (0.16)
$\sum_{j=0}^4 y_{t+j}$	0.66** (0.27)	0.81*** (0.17)	0.88*** (0.17)	1.09*** (0.23)	0.37*** (0.10)
d_t	-0.54* (0.29)	-0.56*** (0.16)	-0.35** (0.17)	-0.33* (0.18)	-0.94*** (0.10)
ρ	0.72*** (0.08)	0.74*** (0.04)	0.73*** (0.03)	0.80*** (0.04)	0.66*** (0.05)
N	162	159	162	162	144

Notes: The table reports estimates of [equation \(1\)](#). The dependent variable is the federal funds rate, r_t . The explanatory variables are the average of CPI inflation over the current and next four quarters, $\sum_{j=0}^4 \pi_{t+j}$, the average of the output gap over the current and next four quarters, $\sum_{j=0}^4 y_{t+j}$, and the public debt-to-GDP ratio, d_t . The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. The variable d_t is rescaled so that one unit corresponds to 10 percentage points. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of [Clarida, Gali, and Gertler \(2000\)](#), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of [Angeletos, Collard, and Dellas \(2020\)](#) and the excess bond-premium shock of [Gilchrist and Zakrajšek \(2012\)](#). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of [Ramey \(2016\)](#), and the news oil-price shocks of [Känzig \(2021\)](#). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B12: Taylor rule estimates using average expected values of inflation and output

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
$\sum_{j=1}^4 \pi_{t+j}^{GB}$	1.39*** (0.23)	1.15*** (0.27)	1.36*** (0.20)	1.33*** (0.20)	1.18*** (0.08)
$\sum_{j=1}^4 y_{t+j}^{GB}$	0.53*** (0.09)	0.46*** (0.09)	0.43*** (0.07)	0.43*** (0.07)	0.54*** (0.03)
d_t	-0.65*** (0.13)	-0.74*** (0.11)	-0.67*** (0.09)	-0.69*** (0.07)	-0.58*** (0.05)
ρ	0.81*** (0.03)	0.86*** (0.02)	0.85*** (0.03)	0.87*** (0.03)	0.81*** (0.01)
N	122	122	122	122	114

Notes: The table reports estimates of [equation \(1\)](#). The dependent variable is the federal funds rate, r_t . The explanatory variables are the average of expected CPI inflation over the next four quarters, $\sum_{j=1}^4 \pi_{t+j}^{GB}$, the average of the expected output gap over the next four quarters, $\sum_{j=1}^4 y_{t+j}^{GB}$, and the public debt-to-GDP ratio, d_t . Expectations data are taken from the Greenbook. The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. The variable d_t is rescaled so that one unit corresponds to 10 percentage points. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of [Clarida, Gali, and Gertler \(2000\)](#), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of [Angeletos, Collard, and Dellas \(2020\)](#) and the excess bond-premium shock of [Gilchrist and Zakrajšek \(2012\)](#). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of [Ramey \(2016\)](#), and the news oil-price shocks of [Känzig \(2021\)](#). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B13: Taylor rule estimates controlling for the realized values of inflation and the output gap three years ahead

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
π_t	1.01*** (0.24)	1.11*** (0.26)	1.79*** (0.45)	1.58*** (0.48)	0.98*** (0.13)
y_t	0.57*** (0.17)	0.68*** (0.15)	0.74*** (0.20)	0.62*** (0.19)	0.58*** (0.14)
d_t	-0.78*** (0.19)	-0.63*** (0.19)	-0.38 (0.30)	-0.43 (0.29)	-0.83*** (0.12)
π_{t+12}	0.40** (0.18)	0.62** (0.25)	-0.57 (0.58)	0.13 (0.68)	0.77*** (0.21)
y_{t+12}	-0.07 (0.18)	0.29 (0.31)	0.69* (0.38)	0.86 (0.57)	-0.23 (0.16)
ρ	0.65*** (0.12)	0.70*** (0.06)	0.71*** (0.06)	0.71*** (0.09)	0.62*** (0.06)
N	162	159	162	162	144

Notes: The table reports estimates of [equation \(2\)](#). The dependent variable is the federal funds rate, r_t . The explanatory variables are CPI inflation, π_t , the output gap, y_t , the public debt-to-GDP ratio, d_t , and the realized values of inflation and the output gap three years ahead, π_{t+12} and y_{t+12} . The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. The variable d_t is rescaled so that one unit corresponds to 10 percentage points. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of [Clarida, Gali, and Gertler \(2000\)](#), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of [Angeletos, Collard, and Dellas \(2020\)](#) and the excess bond-premium shock of [Gilchrist and Zakrajšek \(2012\)](#). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of [Ramey \(2016\)](#), and the news oil-price shocks of [Känzig \(2021\)](#). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B14: Taylor rule estimates controlling for the realized values of inflation and the output gap five years ahead

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
π_t	0.95*** (0.24)	0.76*** (0.20)	0.83*** (0.18)	0.82*** (0.20)	1.15*** (0.11)
y_t	0.54*** (0.17)	0.57*** (0.12)	0.57*** (0.10)	0.68*** (0.14)	0.55*** (0.11)
d_t	-0.89*** (0.19)	-0.89*** (0.15)	-0.91*** (0.15)	-0.90*** (0.15)	-0.80*** (0.10)
π_{t+20}	0.48*** (0.17)	0.60*** (0.20)	0.65* (0.36)	0.41 (0.31)	0.61*** (0.13)
y_{t+20}	-0.05 (0.16)	0.26 (0.17)	0.13 (0.36)	0.24 (0.30)	-0.22** (0.10)
ρ	0.62*** (0.13)	0.58*** (0.08)	0.52*** (0.05)	0.66*** (0.07)	0.62*** (0.04)
N	156	156	156	156	144

Notes: The table reports estimates of [equation \(2\)](#). The dependent variable is the federal funds rate, r_t . The explanatory variables are CPI inflation, π_t , the output gap, y_t , the public debt-to-GDP ratio, d_t , and the realized values of inflation and the output gap five years ahead, π_{t+20} and y_{t+20} . The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. The variable d_t is rescaled so that one unit corresponds to 10 percentage points. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of [Clarida, Gali, and Gertler \(2000\)](#), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of [Angeletos, Collard, and Dellas \(2020\)](#) and the excess bond-premium shock of [Gilchrist and Zakrajšek \(2012\)](#). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of [Ramey \(2016\)](#), and the news oil-price shocks of [Känzig \(2021\)](#). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B15: Taylor rule estimates controlling for the realized values of inflation and the output gap ten years ahead

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
π_t	0.89*** (0.27)	0.86*** (0.32)	0.89*** (0.23)	1.05*** (0.27)	1.08*** (0.12)
y_t	0.38*** (0.14)	0.56*** (0.16)	0.66*** (0.18)	0.75*** (0.26)	0.32*** (0.09)
d_t	-1.17*** (0.26)	-1.14*** (0.29)	-1.14*** (0.23)	-1.02*** (0.27)	-0.98*** (0.10)
π_{t+40}	-0.02 (0.21)	0.20 (0.34)	0.63* (0.33)	0.13 (0.35)	0.02 (0.11)
y_{t+40}	0.15 (0.15)	-0.08 (0.18)	-0.12 (0.16)	-0.39 (0.31)	0.21*** (0.08)
ρ	0.58*** (0.13)	0.71*** (0.06)	0.64*** (0.06)	0.76*** (0.06)	0.61*** (0.04)
N	136	136	136	136	134

Notes: The table reports estimates of [equation \(2\)](#). The dependent variable is the federal funds rate, r_t . The explanatory variables are CPI inflation, π_t , the output gap, y_t , the public debt-to-GDP ratio, d_t , and the realized values of inflation and the output gap ten years ahead, π_{t+40} and y_{t+40} . The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. The variable d_t is rescaled so that one unit corresponds to 10 percentage points. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of [Clarida, Gali, and Gertler \(2000\)](#), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of [Angeletos, Collard, and Dellas \(2020\)](#) and the excess bond-premium shock of [Gilchrist and Zakrajšek \(2012\)](#). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of [Ramey \(2016\)](#), and the news oil-price shocks of [Känzig \(2021\)](#). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B16: Taylor rule estimates controlling for Drechsel (2024)’s number of interactions between the U.S. president and the Federal Reserve chair

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
π_t^{cpi}	1.08*** (0.25)	1.42*** (0.18)	1.35*** (0.20)	1.41*** (0.16)	1.14*** (0.11)
y_t	0.46** (0.19)	0.43*** (0.10)	0.52*** (0.11)	0.42*** (0.11)	0.39*** (0.10)
d_t	-0.99*** (0.24)	-0.84*** (0.12)	-0.83*** (0.18)	-1.00*** (0.15)	-0.98*** (0.09)
# interactions $_t$	-0.24 (0.26)	-1.09*** (0.34)	-0.80*** (0.22)	-1.47*** (0.39)	0.31*** (0.12)
ρ	0.62*** (0.13)	0.58*** (0.06)	0.60*** (0.05)	0.59*** (0.07)	0.63*** (0.05)
N	150	150	150	150	144

Notes: The table reports estimates of equation (2). The dependent variable is the federal funds rate, r_t . The explanatory variables are CPI inflation, π_t , the output gap, y_t , the public debt-to-GDP ratio, d_t , and the number of interactions—phone calls and in-person meetings—between the U.S. president and the Federal Reserve chair, # interactions $_t$, taken from Drechsel (2024). The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. The variable d_t is rescaled so that one unit corresponds to 10 percentage points. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of Clarida, Gali, and Gertler (2000), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of Angeletos, Collard, and Dellas (2020) and the excess bond-premium shock of Gilchrist and Zakrajšek (2012). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of Ramey (2016), and the news oil-price shocks of Känzig (2021). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B17: Taylor rule estimates controlling for the distance to the next election

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
π_t^{cpi}	1.18*** (0.25)	1.15*** (0.22)	1.29*** (0.17)	1.17*** (0.22)	1.19*** (0.12)
y_t	0.61*** (0.20)	0.72*** (0.16)	0.82*** (0.16)	0.84*** (0.17)	0.47*** (0.12)
d_t	-0.74*** (0.22)	-0.77*** (0.19)	-0.56*** (0.16)	-0.62*** (0.19)	-0.91*** (0.12)
Dist. to elec.	0.18 (0.30)	-0.32 (0.35)	-0.17 (0.29)	0.10 (0.17)	0.21** (0.09)
ρ	0.68*** (0.11)	0.75*** (0.05)	0.72*** (0.03)	0.77*** (0.04)	0.65*** (0.05)
N	162	159	162	162	144

Notes: The table reports estimates of [equation \(2\)](#). The dependent variable is the federal funds rate, r_t . The explanatory variables are CPI inflation, π_t , the output gap, y_t , the public debt-to-GDP ratio, d_t , and the number of quarters until the next midterm or presidential election, Dist. to elec. _{t} . The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. The variable d_t is rescaled so that one unit corresponds to 10 percentage points. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of [Clarida, Gali, and Gertler \(2000\)](#), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of [Angeletos, Collard, and Dellas \(2020\)](#) and the excess bond-premium shock of [Gilchrist and Zakrajšek \(2012\)](#). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of [Ramey \(2016\)](#), and the news oil-price shocks of [Känzig \(2021\)](#). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B18: Taylor rule estimates controlling for NBER recession dates

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
π_t	1.40*** (0.26)	1.47*** (0.22)	1.44*** (0.18)	1.55*** (0.24)	1.26*** (0.12)
y_t	0.44** (0.20)	0.55*** (0.15)	0.59*** (0.14)	0.60*** (0.17)	0.40*** (0.08)
d_t	-0.78*** (0.21)	-0.79*** (0.15)	-0.74*** (0.14)	-0.80*** (0.14)	-1.00*** (0.09)
NBER rec. $_t$	-5.49*** (1.66)	-8.13*** (2.73)	-7.12*** (1.77)	-11.00*** (3.29)	-3.73*** (0.88)
ρ	0.72*** (0.10)	0.78*** (0.05)	0.75*** (0.03)	0.81*** (0.04)	0.67*** (0.05)
N	162	159	162	162	144

Notes: The table reports estimates of [equation \(2\)](#). The dependent variable is the federal funds rate, r_t . The explanatory variables are CPI inflation, π_t , the output gap, y_t , the public debt-to-GDP ratio, d_t , and a dummy variable for NBER recession dates, NBER rec. $_t$. The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. The variable d_t is rescaled so that one unit corresponds to 10 percentage points. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of [Clarida, Gali, and Gertler \(2000\)](#), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of [Angeletos, Collard, and Dellas \(2020\)](#) and the excess bond-premium shock of [Gilchrist and Zakrajšek \(2012\)](#). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of [Ramey \(2016\)](#), and the news oil-price shocks of [Känzig \(2021\)](#). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B19: Taylor rule estimates controlling for BAA-AAA spread

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
π_t	1.20*** (0.25)	1.10*** (0.22)	1.26*** (0.18)	1.20*** (0.20)	1.18*** (0.13)
y_t	0.53** (0.24)	0.57*** (0.20)	0.75*** (0.20)	0.79*** (0.19)	0.54*** (0.13)
d_t	-0.77*** (0.23)	-0.86*** (0.17)	-0.64*** (0.17)	-0.62*** (0.16)	-0.90*** (0.11)
Baa-Aaa sp. _t	-0.58 (1.02)	-1.04 (0.79)	-0.49 (0.80)	-0.16 (0.71)	0.67* (0.38)
ρ	0.69*** (0.11)	0.79*** (0.04)	0.74*** (0.03)	0.77*** (0.04)	0.63*** (0.05)
N	162	159	162	162	144

Notes: The table reports estimates of [equation \(2\)](#). The dependent variable is the federal funds rate, r_t . The explanatory variables are CPI inflation, π_t , the output gap, y_t , the public debt-to-GDP ratio, d_t , and the BAA-AAA spread, Baa-Aaa sp._t. The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. The variable d_t is rescaled so that one unit corresponds to 10 percentage points. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of [Clarida, Gali, and Gertler \(2000\)](#), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of [Angeletos, Collard, and Dellas \(2020\)](#) and the excess bond-premium shock of [Gilchrist and Zakrajsek \(2012\)](#). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of [Ramey \(2016\)](#), and the news oil-price shocks of [Känzig \(2021\)](#). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B20: Taylor rule estimates controlling for [Gilchrist and Zakrajšek \(2012\)](#) spread

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
π_t	0.99*** (0.22)	0.88*** (0.19)	0.96*** (0.14)	0.85*** (0.19)	1.03*** (0.10)
y_t	0.48*** (0.17)	0.46*** (0.12)	0.56*** (0.13)	0.57*** (0.14)	0.40*** (0.07)
d_t	-0.77*** (0.18)	-0.86*** (0.13)	-0.74*** (0.12)	-0.76*** (0.14)	-0.96*** (0.07)
gz_t	-0.86*** (0.35)	-0.86*** (0.27)	-0.99*** (0.25)	-1.16*** (0.27)	-0.50*** (0.15)
ρ	0.64*** (0.11)	0.72*** (0.05)	0.67*** (0.04)	0.73*** (0.05)	0.59*** (0.04)
N	162	159	162	162	144

Notes: The table reports estimates of [equation \(2\)](#). The dependent variable is the federal funds rate, r_t . The explanatory variables are CPI inflation, π_t , the output gap, y_t , the public debt-to-GDP ratio, d_t , and the [Gilchrist and Zakrajšek \(2012\)](#) spread, gz_t . The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. The variable d_t is rescaled so that one unit corresponds to 10 percentage points. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of [Clarida, Gali, and Gertler \(2000\)](#), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of [Angeletos, Collard, and Dellas \(2020\)](#) and the excess bond-premium shock of [Gilchrist and Zakrajšek \(2012\)](#). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of [Ramey \(2016\)](#), and the news oil-price shocks of [Känzig \(2021\)](#). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B21: Taylor rule estimates controlling for NFCI

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
π_t	1.33*** (0.32)	1.37*** (0.22)	1.48*** (0.20)	1.44*** (0.21)	1.23*** (0.14)
y_t	0.57*** (0.21)	0.63*** (0.16)	0.68*** (0.15)	0.75*** (0.16)	0.44*** (0.11)
d_t	-0.75*** (0.23)	-0.72*** (0.16)	-0.61*** (0.15)	-0.53*** (0.14)	-0.95*** (0.10)
NFCI _t	-0.54 (0.69)	-0.50 (0.46)	-0.68 (0.45)	-0.38 (0.54)	-0.24 (0.27)
ρ	0.70*** (0.11)	0.73*** (0.05)	0.72*** (0.03)	0.75*** (0.04)	0.64*** (0.04)
N	162	159	162	162	144

Notes: The table reports estimates of [equation \(2\)](#). The dependent variable is the federal funds rate, r_t . The explanatory variables are CPI inflation, π_t , the output gap, y_t , the public debt-to-GDP ratio, d_t , and the Chicago Fed National Financial Conditions Index, NFCI_t. The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. The variable d_t is rescaled so that one unit corresponds to 10 percentage points. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of [Clarida, Gali, and Gertler \(2000\)](#), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of [Angeletos, Collard, and Dellas \(2020\)](#) and the excess bond-premium shock of [Gilchrist and Zakrajšek \(2012\)](#). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of [Ramey \(2016\)](#), and the news oil-price shocks of [Känzig \(2021\)](#). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B22: Taylor rule estimates controlling for the CBOE Volatility Index

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
π_t	0.33 (0.41)	0.97*** (0.36)	0.70*** (0.18)	1.10*** (0.23)	0.63*** (0.09)
y_t	0.91*** (0.25)	1.00*** (0.16)	0.86*** (0.14)	0.74*** (0.13)	0.95*** (0.07)
d_t	-0.51** (0.21)	-0.30** (0.15)	-0.51*** (0.09)	-0.43*** (0.08)	-0.43*** (0.07)
VIX _{<i>t</i>}	-0.16** (0.07)	-0.08 (0.06)	-0.10*** (0.04)	-0.05 (0.04)	-0.09*** (0.02)
ρ	0.89*** (0.02)	0.89*** (0.02)	0.86*** (0.02)	0.89*** (0.02)	0.88*** (0.01)
N	120	113	116	116	104

Notes: The table reports estimates of [equation \(2\)](#). The dependent variable is the federal funds rate, r_t . The explanatory variables are CPI inflation, π_t , the output gap, y_t , the public debt-to-GDP ratio, d_t , and the CBOE Volatility Index, VIX_{*t*}. The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. The variable d_t is rescaled so that one unit corresponds to 10 percentage points. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of [Clarida, Gali, and Gertler \(2000\)](#), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of [Angeletos, Collard, and Dellas \(2020\)](#) and the excess bond-premium shock of [Gilchrist and Zakrajsek \(2012\)](#). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of [Ramey \(2016\)](#), and the news oil-price shocks of [Känzig \(2021\)](#). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B23: Taylor rule estimates controlling for the S&P500 returns

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
π_t	1.18*** (0.25)	1.13*** (0.22)	1.30*** (0.18)	1.23*** (0.22)	1.19*** (0.12)
y_t	0.60*** (0.20)	0.69*** (0.14)	0.81*** (0.13)	0.84*** (0.16)	0.46*** (0.11)
d_t	-0.74*** (0.22)	-0.75*** (0.18)	-0.56*** (0.17)	-0.53*** (0.19)	-0.94*** (0.10)
$\log(\text{S\&P } 500)_t$	-0.72 (4.37)	0.85 (5.50)	2.64 (4.67)	-5.63 (4.50)	-1.09 (1.51)
ρ	0.68*** (0.11)	0.76*** (0.05)	0.71*** (0.03)	0.78*** (0.04)	0.64*** (0.04)
N	162	159	162	162	144

Notes: The table reports estimates of [equation \(2\)](#). The dependent variable is the federal funds rate, r_t . The explanatory variables are CPI inflation, π_t , the output gap, y_t , the public debt-to-GDP ratio, d_t , and the S&P500 returns, $\Delta \log(\text{S\&P } 500)_t$. The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. The variable d_t is rescaled so that one unit corresponds to 10 percentage points. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of [Clarida, Gali, and Gertler \(2000\)](#), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of [Angeletos, Collard, and Dellas \(2020\)](#) and the excess bond-premium shock of [Gilchrist and Zakrajsek \(2012\)](#). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of [Ramey \(2016\)](#), and the news oil-price shocks of [Känzig \(2021\)](#). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

C Stylized New Keynesian model

In this section, we present in detail the economy described in [Section 3](#). We also report robustness checks for the impulse response functions to a liquidity shock.

C.1 Model framework

The model features five agents: final-goods producers, intermediate-goods producers, households, and the fiscal and monetary authorities. The production sectors follow standard New Keynesian formulations. Below we describe the problem faced by each agent.

C.1.1 Final-goods producers

At each point in time t , perfectly competitive firms produce a homogeneous final good Y_t by aggregating a continuum of differentiated intermediate goods $Y_t(i)$, $i \in [0, 1]$, according to $Y_t = \left[\int_0^1 Y_t(i)^{\frac{1}{1+\lambda_p}} di \right]^{1+\lambda_p}$. Profit maximization and the zero-profit condition imply that the price of the final good, P_t , is a CES aggregate of the prices of intermediate goods $P_t(i)$: $P_t = \left[\int_0^1 P_t(i)^{-\frac{1}{\lambda_p}} di \right]^{-\lambda_p}$. The corresponding demand for each intermediate variety i is then $Y_t(i) = \left(\frac{P_t(i)}{P_t} \right)^{-\frac{1+\lambda_p}{\lambda_p}} Y_t$.

C.1.2 Intermediate-goods producers

Each intermediate good i is produced by a monopolistically competitive firm according to $Y_t(i) = L_t(i)^{1-\alpha}$ where $L_t(i)$ denotes the amount of labor employed by firm i . Following [Calvo \(1983\)](#), in each period a fraction ξ_p of firms cannot adjust their price, while the remaining fraction chooses $P_t(i)$ optimally to maximize the expected discounted value of future profits:

$$\mathbb{E}_t \left\{ \sum_{s=0}^{\infty} \xi_p^s \beta^s \frac{\Lambda_{t+s}}{\Lambda_t} [P_t(i) Y_{t+s}(i) - W_{t+s} L_{t+s}(i)] \right\} \quad (7)$$

subject to the demand function and cost-minimization conditions. In this expression, Λ_t denotes the marginal utility of nominal income of the representative household that owns the firm, and W_t is the nominal wage.

C.1.3 Household

The representative household maximizes expected lifetime utility:

$$\mathbb{E}_t \sum_{s=0}^{\infty} \beta^s [\log(C_{t+s}) - L_{t+s} + \phi_B(B_{t+s} - \eta_t B)^2], \quad (8)$$

where C_t denotes aggregate consumption, B_t the real value of bond holdings, B its steady-state value, and η_t a shock to households' preference for holding bonds. We interpret this shock as a liquidity or safety shock, since in equilibrium household bond holdings correspond to government debt, which is both liquid and safe. The liquidity shock follows an AR(1) process:

$$\log(\eta_t) = (1 - \rho_\eta)\eta + \rho_\eta \log(\eta_{t-1}) + \varepsilon_{\eta,t},$$

where η is the steady-state value. The household's nominal flow budget constraint is

$$P_t C_t + P_t B_t + P_t T_t \leq R_{t-1} P_{t-1} B_{t-1} + W_t L_t + \Pi_t,$$

where T_t denotes lump-sum taxes, Π_t profits from firm ownership, and the aggregate consumption index is

$$C_t = \left(\int_0^1 C_t(i)^{\frac{1}{1+\lambda_p}} di \right)^{1+\lambda_p}.$$

C.1.4 The government and market clearing

The monetary authority sets the nominal interest rate according to a feedback rule of the form

$$\frac{R_t}{R} = \left(\frac{R_t^*}{R} \right)^{\phi_{R^*}} \left(\frac{\pi_t}{\pi} \right)^{\phi_\pi} \left(\frac{Y_t}{Y_t^*} \right)^{\phi_y} \quad (9)$$

where R denotes the steady-state gross nominal interest rate, $\pi_t \equiv \frac{P_t}{P_{t-1}}$ is gross inflation, and π its steady-state value. Following the DSGE literature, variables with an asterisk * refer to the natural or flexible-price equilibrium—that is, the allocation that would prevail in the absence of nominal rigidities.

The novelty in [equation \(9\)](#) is the inclusion of the term $\left(\frac{R_t^*}{R} \right)^{\phi_{R^*}}$. This assumption allows the monetary authority to target the natural rate and is important for reconciling our insight that debt is informative about the natural rate, whereas the monetary authority responds to the natural rate rather than to debt itself.

The government's flow budget constraint in real terms is

$$D_t = R_{t-1}\pi_t^{-1}D_{t-1} - T_t, \quad (10)$$

where D_t denotes the real value of government debt and T_t lump-sum taxes. We abstract from government purchases and assume that the government issues only short-term bonds.

Fiscal policy is non-Ricardian because bonds enter the household's utility function. It is therefore necessary to specify the tax rule followed by the fiscal authority, which we assume takes the form

$$T_t - \tilde{T} = \tau_y Y_t + \tau_d D_{t-1} + \tau_R (R_t \pi_{t+1}^{-1}), \quad (11)$$

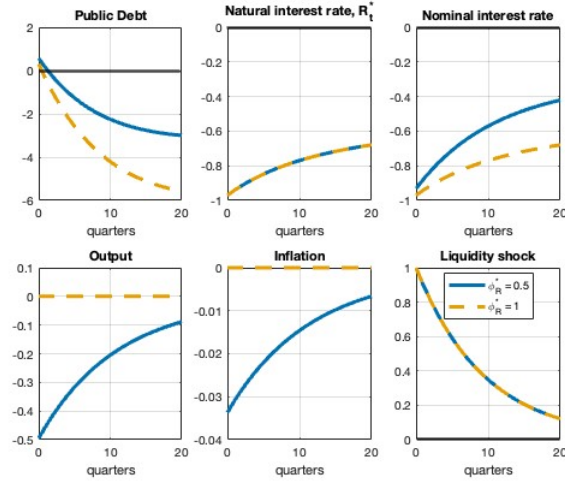
where $\tilde{T}$ is the steady-state level of tax revenues. The parameter τ_y captures the responsiveness of taxes to output fluctuations, τ_d governs how aggressively the government stabilizes debt, and τ_R measures the sensitivity of taxes to the real interest rate. The last term is novel relative to the standard literature and reflects the idea that issuing debt becomes more costly when interest rates rise, prompting the fiscal authority to increase tax revenues.

Finally, the aggregate resource constraint is $C_t = Y_t$, and the bond-market-clearing condition is $B_t = D_t$.

C.2 Impulse response function to a liquidity shock - robustness

Figure C.1 reports the impulse response function to a liquidity shock with $\tau_R = 0.2$.

Figure C.1: Impulse response functions to a liquidity shock - $\tau_R = 0.2$



Notes: The figure shows the impulse response functions to a one-standard-deviation liquidity (safety) shock that raises households' preference for holding government bonds. The blue line corresponds to the Taylor rule with $\phi_{R^*} = 0.5$, while the yellow line corresponds to the rule with $\phi_{R^*} = 1$. The panels display, in order, the responses of public debt, the natural rate of interest, the nominal interest rate, output, inflation, and the liquidity shock itself. The figure uses a small tax response to interest rate changes, $\tau_R = 0.2$.

D Robustness of the estimates on the natural rate

In this section, we show that controlling for other existing estimates of the natural rate does not alter our main finding on the public debt-to-GDP ratio (Table 3). Table D.1 includes the natural rate estimated by Laubach and Williams (2003), while Table D.2 uses the estimate from Holston, Laubach, and Williams (2017). Across nine out of the ten specifications reported in these tables, the coefficient on the public debt-to-GDP remains strongly statistically significant. These results motivate our attempt to construct a measure of the natural rate that explicitly incorporates public debt as a relevant predictor.

Table D.1: Taylor rule using the natural rate estimated by [Laubach and Williams \(2003\)](#)

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
π_t	1.17*** (0.25)	1.18*** (0.19)	1.24*** (0.18)	1.36*** (0.21)	1.18*** (0.12)
y_t	0.63* (0.38)	0.77*** (0.26)	0.98*** (0.31)	1.26*** (0.32)	0.62*** (0.15)
d_t	-0.80 (0.50)	-0.81** (0.40)	-0.85** (0.36)	-1.09*** (0.35)	-1.43*** (0.24)
$R_t^{*,lw}$	-0.18 (1.53)	-0.18 (1.00)	-0.72 (1.13)	-2.00* (1.14)	-1.18* (0.61)
ρ	0.68*** (0.11)	0.74*** (0.05)	0.74*** (0.04)	0.79*** (0.04)	0.63*** (0.04)
N	162	159	162	162	144

Notes: The table reports estimates of [equation \(2\)](#). The dependent variable is the federal funds rate, r_t . The explanatory variables are CPI inflation, π_t , the output gap, y_t , the public debt-to-GDP ratio, d_t , and the natural rate of interest estimated by [Laubach and Williams \(2003\)](#), $R_t^{*,lw}$. The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. The variable d_t is rescaled so that one unit corresponds to 10 percentage points. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of [Clarida, Gali, and Gertler \(2000\)](#), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of [Angeletos, Collard, and Dellas \(2020\)](#) and the excess bond-premium shock of [Gilchrist and Zakrajšek \(2012\)](#). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of [Ramey \(2016\)](#), and the news oil-price shocks of [Känzig \(2021\)](#). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table D.2: Taylor rule using the natural rate estimated by [Holston, Laubach, and Williams \(2017\)](#)

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
π_t	1.17*** (0.25)	1.18*** (0.19)	1.27*** (0.18)	1.34*** (0.21)	1.22*** (0.13)
y_t	0.70* (0.37)	0.91*** (0.30)	1.13*** (0.33)	1.39*** (0.36)	0.56*** (0.16)
d_t	-0.96* (0.57)	-1.07*** (0.40)	-1.13*** (0.40)	-1.46*** (0.48)	-1.17*** (0.22)
$R_t^{*,hlw}$	-0.54 (1.46)	-0.83 (0.99)	-1.46 (1.13)	-2.55* (1.30)	-0.56 (0.53)
ρ	0.68*** (0.11)	0.75*** (0.05)	0.75*** (0.03)	0.79*** (0.04)	0.64*** (0.05)
N	162	159	162	162	144

Notes: The table reports estimates of [equation \(2\)](#). The dependent variable is the federal funds rate, r_t . The explanatory variables are CPI inflation, π_t , the output gap, y_t , the public debt-to-GDP ratio, d_t , and the natural rate of interest estimated by [Holston, Laubach, and Williams \(2017\)](#), $R_t^{*,hlw}$. The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. The variable d_t is rescaled so that one unit corresponds to 10 percentage points. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of [Clarida, Gali, and Gertler \(2000\)](#), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of [Angelesos, Collard, and Dellas \(2020\)](#) and the excess bond-premium shock of [Gilchrist and Zakrajšek \(2012\)](#). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of [Ramey \(2016\)](#), and the news oil-price shocks of [Känzig \(2021\)](#). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

E Time-varying parameter VAR

This section describes the time-varying parameter VAR model ([Appendix E.1](#)) and additional results using our estimated debt-informed measure of the natural rate ([Appendix E.2](#))

E.1 TVP-VAR model

This section describes the time-varying parameter VAR following [Primiceri \(2005\)](#). The model is given by

$$y_t = c_t + B_{1,t}y_{t-1} + \cdots + B_{k,t}y_{t-k} + u_t \quad t = 1, \dots, T$$

where y_t is an $n \times 1$ vector of observable variables, c_t is an $n \times 1$ vector of time-varying coefficients on the constant terms, and $B_{i,t}$ ($i = 1, \dots, k$) are $n \times n$ matrices of time-varying coefficients. The residuals u_t are heteroskedastic structural shocks with variance-covariance matrix Ω_t , such that

$$A_t \Omega_t A_t' = \Sigma_t \Sigma_t',$$

where A_t is a lower triangular matrix

$$A_t = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ \alpha_{21,t} & 1 & \cdots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ \alpha_{n1,t} & \cdots & \alpha_{nn-1,t} & 1 \end{bmatrix},$$

and Σ_t is a diagonal matrix of time-varying standard deviations:

$$\Sigma_t = \begin{bmatrix} \sigma_{1,t} & 0 & \cdots & 0 \\ 0 & \sigma_{2,t} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_{n,t} \end{bmatrix}.$$

The model can therefore be written as

$$y_t = c_t + B_{1,t}y_{t-1} + \cdots + B_{k,t}y_{t-k} + A_t^{-1}\Sigma_t\varepsilon_t, \quad \text{Var}(\varepsilon_t) = I_n$$

where I_n is the $n \times n$ identity matrix. Stacking all coefficients in the vector B_t , the system can be

expressed as

$$\begin{cases} y_t = X_t' B_t + A_t^{-1} \Sigma_t \varepsilon_t, \\ X_t' = I_n \otimes [1, y_{t-1}', \dots, y_{t-k}'], \end{cases}$$

where $\otimes$ denotes the Kronecker product.

The coefficients evolve according to random walks:

$$\begin{cases} B_t = B_{t-1} + \nu_t, \\ \alpha_t = \alpha_{t-1} + \zeta_t, \\ \log \sigma_t = \log \sigma_{t-1} + \eta_t, \end{cases}$$

where ν_t , ζ_t , and η_t are mutually uncorrelated innovations.

The joint variance–covariance matrix of the innovations is block-diagonal:

$$V = \text{Var} \left(\begin{bmatrix} \varepsilon_t \\ \nu_t \\ \zeta_t \\ \eta_t \end{bmatrix} \right) = \begin{bmatrix} I_n & 0 & 0 & 0 \\ 0 & Q & 0 & 0 \\ 0 & 0 & S & 0 \\ 0 & 0 & 0 & W \end{bmatrix},$$

where S is block diagonal, with each block corresponding to the contemporaneous parameters of a given equation (so that these coefficients evolve independently across equations).

Following [Primiceri \(2005\)](#), the priors are specified as:

$$\begin{aligned} B_0 &\sim \mathcal{N}(\hat{B}_{OLS}, 4 \cdot \text{Var}(\hat{B}_{OLS})), \\ A_0 &\sim \mathcal{N}(\hat{A}_{OLS}, 4 \cdot \text{Var}(\hat{A}_{OLS})), \\ \log \sigma_0 &\sim \mathcal{N}(\log \hat{\sigma}_{OLS}, I_n), \\ Q &\sim \mathcal{IW}(k_Q^2 \cdot 40 \cdot \text{Var}(\hat{B}_{OLS}), 40), \\ W &\sim \mathcal{IW}(k_W^2 \cdot 4 \cdot I_n, 5), \\ S_1 &\sim \mathcal{IW}(k_S^2 \cdot 2 \cdot \text{Var}(\hat{A}_{1,OLS}), 2), \\ S_2 &\sim \mathcal{IW}(k_S^2 \cdot 3 \cdot \text{Var}(\hat{A}_{2,OLS}), 3), \\ S_3 &\sim \mathcal{IW}(k_S^2 \cdot 4 \cdot \text{Var}(\hat{A}_{3,OLS}), 4) \end{aligned}$$

where $\hat{B}_{OLS}$, $\hat{A}_{OLS}$, and $\hat{\sigma}_{OLS}$ are the OLS estimates. S_1 , S_2 , and S_3 denote the corresponding blocks of S , and $\hat{A}_{i,OLS}$ the analogous blocks of $\hat{A}_{OLS}$. As in [Primiceri \(2005\)](#), we set $k_Q = k_W = 0.01$ and $k_S = 0.1$. We use a ten-year training sample starting in 1961:Q1 and include only one lag ($p = 1$), given the large number of parameters induced by the inclusion of an additional variable.

The model features three sources of time variation: (i) the time variation in the lag coefficients of the model (captured by the innovation term ν_t in the random walk process for B_t), (ii) the time variation in the simultaneous response of variable j to variable i (captured by the innovation term ζ_t in the random walk process α_t), and (iii) the time variation in the variances of the shocks (captured by the innovation term η_t in the geometric random walk process for σ_t).

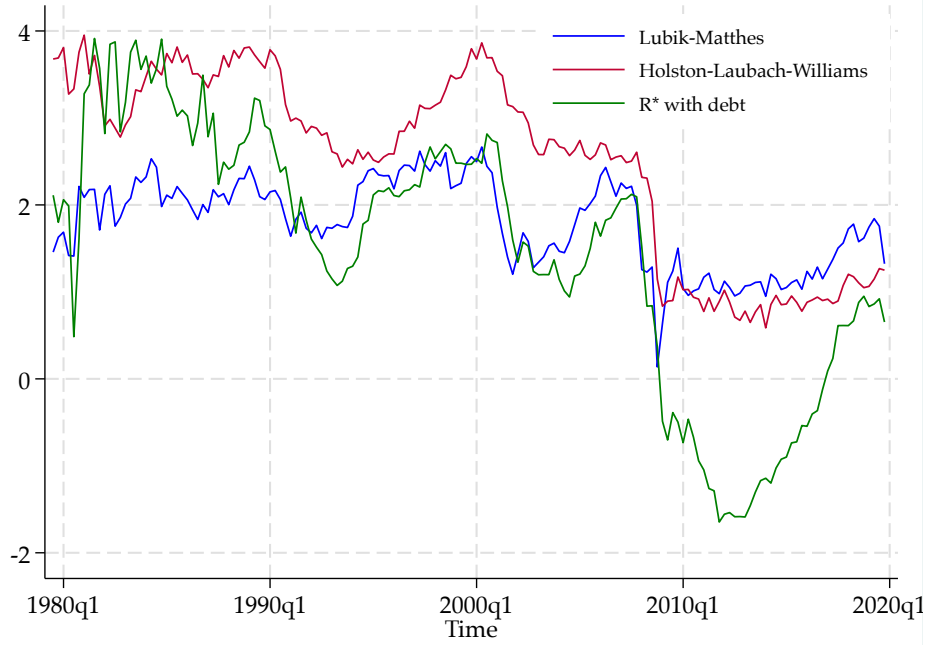
The model is estimated using Bayesian methods via a Gibbs sampler, a specific type of Markov Chain Monte Carlo (MCMC) algorithm. We run 10,000 iterations, discarding the first 2000 as burn-in. Conditional forecasts are obtained by keeping coefficients fixed at each point in time.³⁰

E.2 TVP-VAR results

[Figure E.1](#) shows our estimate of the natural rate and compares it to the measure of [Lubik and Matthes \(2015\)](#) and [Holston, Laubach, and Williams \(2017\)](#). The main difference is that our measure becomes significantly negative during the zero lower bound period.

³⁰Allowing the coefficients to vary over time yields a correlation of about 99% with our baseline measure.

Figure E.1: Different measures of the natural rate of interest



Notes: The figure shows in green our estimates of the natural rate, in blue those of [Lubik and Matthes \(2015\)](#), and in red those of [Holston, Laubach, and Williams \(2017\)](#).

Equipped with our debt-informed measure of the natural rate, we show in [Table 3](#) that it explains the finding reported in [Table 1](#). We now demonstrate that this result is robust to several alternative specifications.

First, we construct our debt-informed measure of the natural rate, $R^{*,d}$, using PCE inflation, following [Lubik and Matthes \(2015\)](#). Second, we estimate the TVP-VAR using the nominal interest rate instead of the real rate. We then recover the real rate as the difference between the five-year ahead forecasts of the nominal rate and inflation. Third, we re-estimate the TVP-VAR with two lags.

The results are reported in [Table E.1](#), [Table E.2](#), and [Table E.3](#), respectively. Across specifications, the results are stable in terms of sign and statistical significance: the coefficient on public debt-over-GDP ratio becomes insignificant in most cases and is fully absorbed by our debt-informed measure of the natural rate. Estimating the TVP-VAR with the nominal rate rather than the real rate affects the magnitude of the coefficient on our debt-informed measure of the natural rate, which is somewhat larger in this specification.

We replicate the same robustness checks for [Table 4](#). The results show that our debt-informed

measure of the natural rate is a predictor of the Federal Reserve policy rate. Results are stable across the different exercises (Table E.4, Table E.5, and Table E.6).

Table E.1: Taylor rule estimates using our debt-informed natural rate measure and PCE inflation

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
π_t^{pce}	0.99*** (0.29)	1.18*** (0.21)	1.10*** (0.21)	1.10*** (0.17)	1.23*** (0.23)
y_t	0.27 (0.19)	0.35** (0.15)	0.30** (0.14)	0.31** (0.13)	0.38** (0.16)
d_t	-0.14 (0.15)	0.01 (0.09)	-0.05 (0.07)	-0.02 (0.08)	-0.36 (0.29)
$R_t^{*,d}$	0.95*** (0.27)	0.96*** (0.18)	0.97*** (0.16)	1.00*** (0.16)	0.63* (0.37)
ρ	0.41 (0.27)	0.46*** (0.12)	0.43*** (0.12)	0.43*** (0.12)	0.56*** (0.11)
N	162	159	162	162	144

Notes: The table reports estimates of equation (2). The dependent variable is the federal funds rate, r_t . The explanatory variables are PCE inflation, π_t , the output gap, y_t , the public debt-to-GDP ratio, d_t , and our estimate of the natural rate of interest, $R_t^{*,d}$, constructed using PCE inflation instead of CPI inflation in the TVP-VAR. The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. The variable d_t is rescaled so that one unit corresponds to 10 percentage points. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of Clarida, Gali, and Gertler (2000), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of Angeletos, Collard, and Dellas (2020) and the excess bond-premium shock of Gilchrist and Zakrajšek (2012). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of Ramey (2016), and the news oil-price shocks of Känzig (2021). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table E.2: Taylor rule estimates based on our debt-informed natural rate measure constructed with the nominal rate in the TVP-VAR

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
π_t^{cpi}	0.48*** (0.13)	0.44*** (0.07)	0.48*** (0.06)	0.47*** (0.10)	0.35*** (0.06)
y_t	0.14 (0.09)	0.09** (0.05)	0.12** (0.06)	0.15* (0.09)	0.08*** (0.03)
d_t	0.12 (0.13)	0.20*** (0.08)	0.13 (0.09)	0.08 (0.11)	0.28*** (0.11)
$R_t^{*,d}$	1.75*** (0.22)	1.85*** (0.11)	1.75*** (0.13)	1.67*** (0.20)	1.97*** (0.15)
ρ	0.11 (0.37)	-0.04 (0.19)	-0.01 (0.16)	0.22 (0.19)	-0.16 (0.12)
N	162	159	162	162	144

Notes: The table reports estimates of [equation \(2\)](#). The dependent variable is the federal funds rate, r_t . The explanatory variables are PCE inflation, π_t , the output gap, y_t , the public debt-to-GDP ratio, d_t , and our estimate of the natural rate of interest, $R_t^{*,d}$, constructed using the nominal interest rate instead of the real rate in the TVP-VAR. The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. The variable d_t is rescaled so that one unit corresponds to 10 percentage points. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of [Clarida, Gali, and Gertler \(2000\)](#), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of [Angeletos, Collard, and Dellas \(2020\)](#) and the excess bond-premium shock of [Gilchrist and Zakrajšek \(2012\)](#). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of [Ramey \(2016\)](#), and the news oil-price shocks of [Känzig \(2021\)](#). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table E.3: Taylor rule estimates using our debt-informed natural rate measure constructed with two lags in the TVP-VAR

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
π_t	1.19*** (0.19)	1.17*** (0.14)	1.21*** (0.10)	1.15*** (0.12)	1.19*** (0.11)
y_t	0.27 (0.18)	0.20** (0.10)	0.26** (0.11)	0.24** (0.11)	0.36*** (0.13)
d_t	-0.04 (0.22)	0.18 (0.12)	0.06 (0.14)	0.12 (0.13)	-0.52 (0.32)
$R_t^{*,d}$	1.27*** (0.33)	1.57*** (0.18)	1.38*** (0.21)	1.48*** (0.21)	0.54 (0.40)
ρ	0.59*** (0.13)	0.56*** (0.07)	0.55*** (0.04)	0.61*** (0.06)	0.61*** (0.05)
N	162	159	162	162	144

Notes: The table reports estimates of [equation \(2\)](#). The dependent variable is the federal funds rate, r_t . The explanatory variables are PCE inflation, π_t , the output gap, y_t , the public debt-to-GDP ratio, d_t , and our estimate of the natural rate of interest, $R_t^{*,d}$, constructed using two lags instead of one lag in the TVP-VAR. The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. The variable d_t is rescaled so that one unit corresponds to 10 percentage points. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of [Clarida, Gali, and Gertler \(2000\)](#), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of [Angeletos, Collard, and Dellas \(2020\)](#) and the excess bond-premium shock of [Gilchrist and Zakrajšek \(2012\)](#). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of [Ramey \(2016\)](#), and the news oil-price shocks of [Känzig \(2021\)](#). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table E.4: Taylor rule estimates excluding public debt and using our debt-informed natural rate measure with PCE inflation

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
π_t	0.97*** (0.26)	1.23*** (0.23)	1.09*** (0.19)	1.13*** (0.20)	1.00*** (0.13)
y_t	0.23 (0.17)	0.35** (0.16)	0.29** (0.14)	0.34** (0.15)	0.21*** (0.08)
$R_t^{*,d}$	1.10*** (0.19)	0.95*** (0.17)	1.05*** (0.13)	0.99*** (0.15)	1.08*** (0.08)
ρ	0.38 (0.28)	0.48*** (0.12)	0.39*** (0.12)	0.45*** (0.12)	0.43*** (0.08)
N	162	159	162	162	144

Notes: The table reports estimates of [equation \(1\)](#), where the debt-to-GDP ratio is replaced with our debt-informed measure of the natural rate of interest, $R_t^{*,d}$, constructed using PCE inflation instead of CPI inflation in the TVP-VAR. The dependent variable is the federal funds rate, r_t . The explanatory variables are CPI inflation, π_t , the output gap, y_t , and our estimate of the natural rate of interest, $R_t^{*,d}$. The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of [Clarida, Gali, and Gertler \(2000\)](#), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of [Angeletos, Collard, and Dellas \(2020\)](#), and the excess bond-premium shock of [Gilchrist and Zakrajšek \(2012\)](#). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of [Ramey \(2016\)](#), and the news oil-price shocks of [Känzig \(2021\)](#). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table E.5: Taylor rule estimates excluding public debt and using our debt-informed natural rate measure constructed with the nominal rate in the TVP-VAR

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
π_t	0.52*** (0.14)	0.49*** (0.10)	0.53*** (0.08)	0.51*** (0.11)	0.49*** (0.06)
y_t	0.17* (0.10)	0.17** (0.07)	0.18*** (0.06)	0.20*** (0.08)	0.13*** (0.04)
$R_t^{*,d}$	1.58*** (0.15)	1.55*** (0.10)	1.54*** (0.09)	1.52*** (0.12)	1.59*** (0.05)
ρ	0.19 (0.33)	0.22 (0.16)	0.14 (0.13)	0.33** (0.14)	0.09 (0.08)
N	162	159	162	162	144

Notes: The table reports estimates of [equation \(1\)](#), where the debt-to-GDP ratio is replaced with our debt-informed measure of the natural rate of interest, $R_t^{*,d}$, constructed using the nominal interest rate instead of the real rate in the TVP-VAR. The dependent variable is the federal funds rate, r_t . The explanatory variables are CPI inflation, π_t , the output gap, y_t , and our estimate of the natural rate of interest, $R_t^{*,d}$. The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1-\rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of [Clarida, Gali, and Gertler \(2000\)](#), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of [Angeletos, Collard, and Dellas \(2020\)](#), and the excess bond-premium shock of [Gilchrist and Zakrajšek \(2012\)](#). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of [Ramey \(2016\)](#), and the news oil-price shocks of [Känzig \(2021\)](#). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table E.6: Taylor rule estimates excluding public debt and using our debt-informed natural rate measure constructed with two lags in the TVP-VAR

	(1) OLS	(2) GMM	(3) GMM BC	(4) GMM EBP	(5) GMM Shocks
π_t	1.20*** (0.17)	1.10*** (0.12)	1.19*** (0.07)	1.11*** (0.11)	1.24*** (0.09)
y_t	0.27 (0.17)	0.24** (0.10)	0.29*** (0.10)	0.28*** (0.11)	0.26** (0.11)
$R_t^{*,d}$	1.32*** (0.22)	1.35*** (0.13)	1.30*** (0.12)	1.32*** (0.14)	1.14*** (0.10)
ρ	0.59*** (0.13)	0.57*** (0.07)	0.55*** (0.05)	0.62*** (0.06)	0.60*** (0.04)
N	162	159	162	162	144

Notes: The table reports estimates of [equation \(1\)](#), where the debt-to-GDP ratio is replaced with our debt-informed measure of the natural rate of interest, $R_t^{*,d}$, constructed using two lags instead of one lag in the TVP-VAR. The dependent variable is the federal funds rate, r_t . The explanatory variables are CPI inflation, π_t , the output gap, y_t , and our estimate of the natural rate of interest, $R_t^{*,d}$. The specification also includes two lags of the policy rate, r_{t-1} and r_{t-2} , with coefficients ρ_1 and ρ_2 . All coefficients are rescaled by $1 - \rho$, where $\rho \equiv \rho_1 + \rho_2$, so that they represent long-run effects. Column (1) reports OLS estimates. Column (2) reports GMM estimates using the instrument set of [Clarida, Gali, and Gertler \(2000\)](#), which includes lagged variables. Columns (3) and (4) use four lags of the explanatory variables as instruments and additionally include, respectively, the main business-cycle shock of [Angeletos, Collard, and Dellas \(2020\)](#), and the excess bond-premium shock of [Gilchrist and Zakrajšek \(2012\)](#). Column (5) uses only lagged macroeconomic shocks as instruments: the business-cycle shock, the excess bond-premium shock, the government-spending shock of [Ramey \(2016\)](#), and the news oil-price shocks of [Känzig \(2021\)](#). In columns (3)–(5), all shocks are used as instruments with lags 5 through 20. Newey–West standard errors with four lags are reported. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

F DSGE model

In this section, we discuss in detail the model described in [Section 5](#). [Appendix F.1](#) discusses the problems of final good producers, intermediate firms, and employment agencies. [Appendix F.2](#) describes the data used to estimate the model. [Appendix F.3](#) reports the prior and posterior estimates. Finally, [Appendix F.4](#) presents additional results.

F.1 Decision problems and market-clearing conditions

We discuss the the problems of final good producers, intermediate firms, and employment agencies, as well as the market-clearing conditions, of the quantitative model discussed in [Section 5](#).

F.1.1 Final goods producers

At each point in time t , the final consumption good Y_t is produced by a representative, perfectly competitive firm that aggregates a continuum of intermediate goods $Y_t(i)$, $i \in [0, 1]$, according to a CES technology:

$$Y_t = \left(\int_0^1 Y_t(i)^{\frac{1}{1+\lambda_{p,t}}} di \right)^{1+\lambda_{p,t}}$$

The elasticity parameter $\lambda_{p,t}$ evolves according to the exogenous stochastic process

$$\log(1 + \lambda_{p,t}) = (1 - \rho_p) \log(1 + \lambda_p) + \rho_p \log(1 + \lambda_{p,t-1}) + \epsilon_{p,t} - \theta_p \epsilon_{p,t-1}.$$

where $\epsilon_{p,t} \stackrel{i.i.d.}{\sim} \mathcal{N}(0, \sigma_p^2)$ represents a price-markup shock, since $\lambda_{p,t}$ captures the desired markup of prices over marginal cost for intermediate firms. The ARMA(1,1) structure allows the model to capture the moving-average, high-frequency component of inflation.

Each firm chooses $Y_t(i)_{i \in [0,1]}$ to minimize total cost, taking as given the prices $P_t(i)$ of intermediate goods, subject to the aggregation constraint above. The resulting demand for intermediate good i is

$$Y_t(i) = \left(\frac{P_t(i)}{P_t} \right)^{-\frac{1+\lambda_{p,t}}{\lambda_{p,t}}} Y_t,$$

where P_t is the final-goods price index:

$$P_t = \left(\int_0^1 P_t(i)^{-\frac{1}{\lambda_{p,t}}} di \right)^{-\lambda_{p,t}}. \quad (12)$$

F.1.2 Intermediate firms

Each intermediate-good producer i operates under monopolistic competition and produces output according to

$$Y_t(i) = \max\{A_t^{1-\alpha} K_t(i)^\alpha L_t(i)^{1-\alpha} - A_t F, 0\}.$$

where $K_t(i)$ and $L_t(i)$ denote the amounts of capital and labor employed by firm i . The parameter F represents a fixed cost of production, chosen so that profits are zero in the steady state (Christiano, Eichenbaum, and Evans, 2005).

The aggregate technology level A_t follows the stochastic process

$$\Delta \log(A_t) \equiv z_t = (1 - \rho_z)\gamma + \rho_z z_{t-1} + \epsilon_{z,t}, \quad \epsilon_{z,t} \stackrel{i.i.d.}{\sim} \mathcal{N}(0, \sigma_z^2)$$

where $\epsilon_{z,t}$ is a neutral technology shock. Technology is non-stationary, and its growth rate z_t evolves as an AR(1) process.

As in Calvo (1983), in each period a fraction ξ_p of firms cannot reoptimize their prices and instead follows the indexation rule

$$P_t(i) = \pi_{t-1}^{\iota_p} \pi^{1-\iota_p},$$

where $\pi_t \equiv \frac{P_t}{P_{t-1}}$ denotes gross inflation and π its steady-state value. The remaining fraction of firms reoptimizes prices to maximize the expected discounted value of future profits:

$$E_t \left\{ \sum_{s=0}^{\infty} \frac{\beta^s \Lambda_{t+s}}{\Lambda_t} \left[P_t(i) \left(\prod_{k=1}^s \pi_{t+k-1}^{\iota_p} \pi^{1-\iota_p} \right) Y_{t+s}(i) - W_{t+s} L_{t+s}(i) - r_{t+s}^k K_{t+s}(i) \right] \right\}$$

subject to the demand function in equation (12) and the firm's cost-minimization conditions. Here, Λ_t denotes the marginal utility of nominal income for the representative household that owns the firm, and W_t and r_t^k are the nominal wage and rental rate of capital, respectively.

F.1.3 Employment agencies

Firms are owned by a continuum of households indexed by $j \in [0, 1]$. Each household is a monopolistic supplier of specialized labor $L_t(j)$, as in Erceg, Henderson, and Levin (2000). A large number of competitive *employment agencies* aggregate this differentiated labor into a homogeneous labor

input sold to intermediate firms, according to

$$L_t = \left[\int_0^1 L_t(j)^{1/(1+\lambda_{w,t})} dj \right]^{1+\lambda_{w,t}}.$$

As in the case of the final good, the desired wage markup $\lambda_{w,t}$ —the markup of the nominal wage over the household’s marginal rate of substitution—follows the exogenous stochastic process

$$\log(1 + \lambda_{w,t}) = (1 - \rho_w) \log(1 + \lambda_w) + \rho_w \log(1 + \lambda_{w,t-1}) + \varepsilon_{w,t} - \theta_w \varepsilon_{w,t-1},$$

where $\varepsilon_{w,t} \stackrel{i.i.d.}{\sim} \mathcal{N}(0, \sigma_w^2)$ represents the wage-markup shock.

Each perfectly competitive employment agency chooses L_t and $L_t(j)_{j \in [0,1]}$ to maximize profits, is:

$$\max_{L_t, L_t(j)} W_t L_t - \int W_t(j) L_t(j) dj \quad \text{s.t.} \quad L_t = \left[\int_0^1 L_t(j)^{1/(1+\lambda_{w,t})} dj \right]^{1+\lambda_{w,t}}$$

The first-order conditions yield the labor demand for each type of worker:

$$L_t(j) = \left(\frac{W_t(j)}{W_t} \right)^{-(1+\lambda_{w,t})/\lambda_{w,t}} L_t,$$

where $W_t(j)$ denotes the nominal wage received by household j from the employment agencies. Imposing the zero-profit condition and substituting the labor demand yields the aggregate wage index:

$$W_t = \left[\int_0^1 W_t(j)^{-1/\lambda_{w,t}} dj \right]^{-\lambda_{w,t}}.$$

F.1.4 Market-clearing conditions

The aggregate resource constraint must hold in every period:

$$C_t + I_t + G_t + a(u_t) \bar{K}_{t-1} = Y_t,$$

which follows from combining the budget constraints of households and the fiscal authority with the zero-profit conditions of final-good producers and employment agencies. In addition, the market for government bonds clears each period, so that households’ demand for bonds equals the public debt issued by the government:

$$B_t = D_t.$$

F.2 Data and observable variables

All the data used in the model of [Section 5](#) are downloaded from [FRED \(Federal Reserve Economic Data\)](#). The data are quarterly and span the period from 1979:Q3 to 2019:Q4, consistent with the empirical analyses in [Section 2](#) and [Section 4](#).

The observable variables are constructed following [Justiniano, Primiceri, and Tambalotti \(2010\)](#). Real GDP is obtained by dividing nominal GDP by both the population (CNP16OV) and the GDP deflator (GDPDEF). Real consumption and real investment are constructed analogously. Consumption includes personal consumption expenditures on nondurable goods and services, while investment is defined as the sum of consumption expenditures on durables (CD) and private investment. The real wage is measured as hourly compensation in the nonfarm business sector for all workers (COMPNFB) divided by the GDP deflator. Hours worked are computed as total employment (CE160V) multiplied by average weekly hours of production workers (AWHNONAG), divided by the population. Inflation is measured as the quarterly log difference of the GDP deflator, and the nominal interest rate corresponds to the effective federal funds rate. Public debt is measured as the market value of gross federal debt (MVGFD027MNFRBDAL).

The observable variables, expressed in percentage terms, include the log differences of GDP, consumption, investment, and real wages, as well as inflation, the federal funds rate, hours worked, and the log of the public debt-to-GDP ratio. These variables are mapped to their model counterparts in the linearized system as follows:

$$\Delta(\log X_t^D) = \Delta \log \left(\frac{X_t^D}{A_t} \right) + \Delta \log(A_t) = 100 * (\hat{x}_t - \hat{x}_{t-1} + \hat{z}_t) + 100\gamma$$

$$\Delta(\log C_t^D) = \Delta \log \left(\frac{C_t^D}{A_t} \right) + \Delta \log(A_t) = 100 * (\hat{c}_t - \hat{c}_{t-1} + \hat{z}_t) + 100\gamma$$

$$\Delta(\log I_t^D) = \Delta \log \left(\frac{I_t^D}{A_t} \right) + \Delta \log(A_t) = 100 * (\hat{i}_t - \hat{i}_{t-1} + \hat{z}_t) + 100\gamma$$

$$\log L_t^D = 100 * \hat{L}_t + 100 \log(L_{ss})$$

$$\Delta \log \left(\frac{W}{P} \right)_t^D = 100 * (\hat{w}_t - \hat{w}_{t-1} + \hat{z}_t) + 100\gamma$$

$$\pi_t^D = 100 * \hat{\pi}_t + \pi_{ss}^o$$

$$R_t^D / 4 = 100 * \hat{R}_t + R_{ss}^o$$

$$\log \left(\frac{B_t}{Y_t} \right)_t^D = 100 * (\hat{d}_t - \hat{y}_t) + 100 \log(d_{ss}) + \varepsilon_t^{me}$$

where variables with a hat denote deviations from the steady state and ε_t^{me} represents a measurement error. The measurement error is included because the debt-to-GDP ratio is detrended using a band-pass filter that removes fluctuations with periodicities longer than 15 years.

F.3 Prior and posterior estimates

In this section, we present the prior and posterior estimates of the model (Table F.1). The priors align with those used by Justiniano, Primiceri, and Tambalotti (2010).

Table F.1: Priors densities and posterior estimates for the baseline model

Coefficient	Description	Prior			Posterior		
		Density ¹	Mean	Std	Mean	Std	[5 , 95] ²
α	Capital share	N	0.30	0.05	0.16	0.02	[0.13, 0.19]
ι_p	Price indexation	B	0.50	0.15	0.32	0.10	[0.16, 0.48]
ι_w	Wage indexation	B	0.50	0.15	0.36	0.09	[0.22, 0.50]
γ	SS technology growth rate	N	0.50	0.03	0.49	0.03	[0.45, 0.53]
h	Consumption habit	B	0.50	0.10	0.57	0.06	[0.48, 0.67]
λ_p^{ss}	SS price markup	N	0.15	0.05	0.19	0.04	[0.12, 0.25]
λ_w^{ss}	SS wage markup	N	0.15	0.05	0.13	0.05	[0.04, 0.21]
$\log(L^{ss})$	SS log-hours	N	0.00	0.50	0.51	0.40	[-0.15, 1.15]
$100(\pi^{ss} - 1)$	SS quarterly inflation	N	0.50	0.10	0.57	0.07	[0.45, 0.69]
$100(\beta^{-1} - 1)$	Discount factor	G	0.25	0.20	0.14	0.10	[0.00, 0.28]
ν	Inverse Frisch elasticity	G	2.00	0.75	3.77	0.95	[2.22, 5.30]
ξ_p	Calvo price	B	0.66	0.10	0.96	0.01	[0.95, 0.98]
ξ_w	Calvo wage	B	0.66	0.10	0.76	0.06	[0.66, 0.86]
χ	Elasticity of capital utilization costs	G	5.00	1.00	5.43	1.01	[3.74, 7.00]

(Continued on next page)

Coefficient	Description	Prior			Posterior		
		Density ¹	Mean	Std	Mean	Std	[5 , 95] ²
S''	Investment adjustment costs	G	4.00	1.00	3.20	0.75	[1.96, 4.36]
ϕ_π	Taylor rule inflation	N	1.70	0.30	1.52	0.26	[1.09, 1.94]
ϕ_x	Taylor rule output	N	0.13	0.05	0.04	0.04	[-0.03, 0.10]
$\phi_{\Delta x}$	Taylor rule output growth	N	0.13	0.05	0.18	0.04	[0.12, 0.24]
ϕ_r	Taylor rule smoothing	B	0.60	0.10	0.83	0.04	[0.76, 0.89]
ϕ_{R^*}	Taylor rule natural rate	N	1.00	0.15	0.80	0.17	[0.52, 1.06]
ϕ_{tax}	Tax rule smoothing	B	0.60	0.20	0.05	0.02	[0.02, 0.09]
ρ_η	Liquidity	B	0.60	0.20	0.96	0.02	[0.93, 0.99]
ρ_{mps}	Monetary policy	B	0.40	0.20	0.12	0.06	[0.02, 0.20]
ρ_z	Neutral technology growth	B	0.60	0.20	0.08	0.04	[0.02, 0.14]
ρ_μ	Investment	B	0.60	0.20	0.24	0.15	[0.03, 0.44]
ρ_p	Price-markup	B	0.60	0.20	0.84	0.05	[0.76, 0.92]
ρ_w	Wage-markup	B	0.60	0.20	0.81	0.15	[0.58, 0.98]
θ_p	Price markup MA	B	0.50	0.20	0.78	0.07	[0.68, 0.88]
θ_w	Wage markup MA	B	0.50	0.20	0.80	0.15	[0.56, 0.97]
ρ_b	Intertemporal preference	B	0.60	0.20	0.78	0.12	[0.67, 0.90]
ρ_g	Government spending	B	0.60	0.20	0.97	0.01	[0.96, 0.99]
ρ_{tax}	Tax	B	0.40	0.20	0.93	0.02	[0.90, 0.97]
ϕ_B	Elasticity of bonds' utility	G	0.25	0.05	0.11	0.03	[0.07, 0.16]
τ_d	Tax rule debt	N	0.15	0.10	0.27	0.04	[0.19, 0.34]
τ_R	Tax rule spending	G	1.00	0.40	1.25	0.25	[0.84, 1.64]
τ_y	Tax rule output	G	0.33	0.10	0.29	0.07	[0.17, 0.41]
τ_{yy}	Tax rule lagged output	G	0.33	0.10	0.13	0.04	[0.07, 0.18]

(Continued on next page)

Coefficient	Description	Prior			Posterior		
		Density ¹	Mean	Std	Mean	Std	[5 , 95] ²
d_y	SS debt-to-GDP	N	0.65	0.50	0.65	0.01	[0.63, 0.67]
$100\sigma_{mps}$	Monetary policy	I	0.10	1.00	0.20	0.01	[0.17, 0.22]
$100\sigma_\eta$	Liquidity	I	0.50	1.00	0.42	0.08	[0.30, 0.54]
$100\sigma_z$	Neutral technology growth	I	0.50	1.00	0.68	0.04	[0.60, 0.75]
$100\sigma_b$	Intertemporal preference	I	0.10	1.00	0.03	0.01	[0.02, 0.05]
$100\sigma_g$	Government spending	I	0.50	1.00	0.55	0.03	[0.50, 0.60]
$100\sigma_\mu$	Investment	I	0.50	1.00	4.82	1.29	[2.77, 6.86]
$100\sigma_{tax}$	Tax	I	0.50	1.00	0.72	0.10	[0.56, 0.88]
$100\sigma_{\lambda_p}$	Price-markup	I	0.10	1.00	0.11	0.01	[0.09, 0.13]
$100\sigma_{\lambda_w}$	Wage-markup	I	0.10	1.00	0.39	0.03	[0.34, 0.44]
$100\sigma_{me}^{debt}$	Measurement error debt	I	1.00	1.00	1.15	0.11	[0.96, 1.33]

¹ Prior distribution type: N = Normal, B = Beta, G = Gamma, I = Inverted-Gamma1.

² Posterior medians and percentiles are computed from three Metropolis–Hastings chains of 120,000 draws each, discarding the first 25,000 as burn-in.

F.4 Model additional results

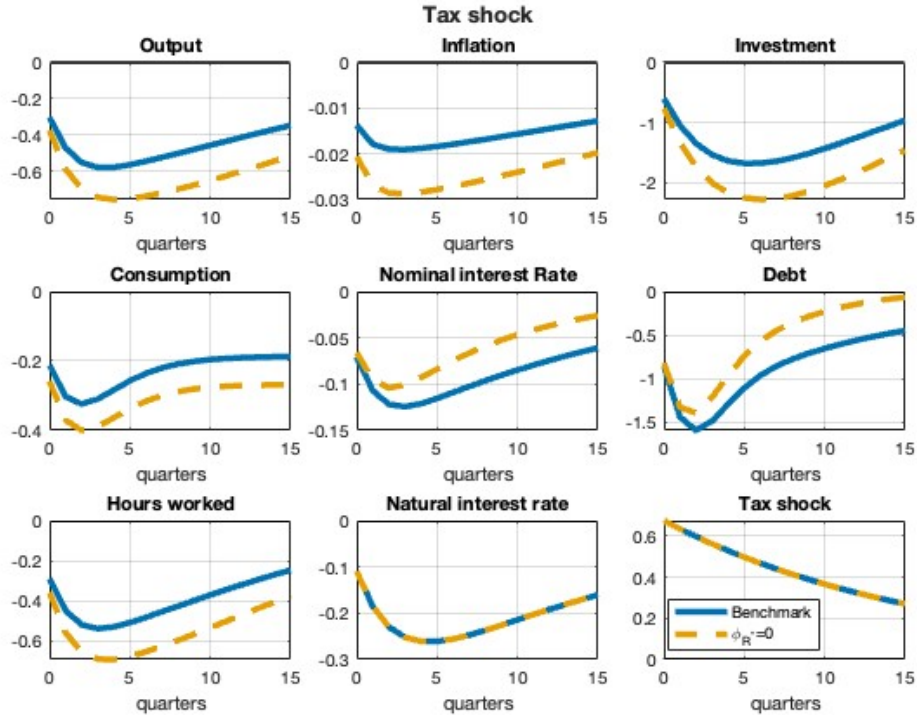
In this section, we report additional results for the model described in [Section 5](#). [Table F.2](#) reports the variance decomposition for all shocks in the model. [Figure F.1](#) reports the impulse response functions to a tax shock.

Table F.2: Posterior variance decomposition at the business cycle frequency

	Preference	Liquidity	Govt spending	Price markup	Wage markup	Monetary policy	Investment	Tax	TFP
GDP	0.05	0.34	0.04	0.01	0.00	0.05	0.04	0.23	0.24
Consumption	0.27	0.14	0.21	0.00	0.01	0.03	0.01	0.09	0.24
Investment	0.10	0.31	0.09	0.02	0.01	0.04	0.18	0.21	0.05
Hours worked	0.06	0.44	0.05	0.01	0.00	0.07	0.04	0.30	0.04
Wage	0.00	0.06	0.01	0.06	0.72	0.00	0.00	0.04	0.11
Inflation	0.00	0.12	0.02	0.80	0.04	0.00	0.00	0.02	0.00
Debt	0.06	0.45	0.09	0.02	0.00	0.10	0.04	0.18	0.05
Interest rate	0.03	0.48	0.04	0.04	0.00	0.24	0.00	0.17	0.00
Natural rate	0.14	0.41	0.10	0.00	0.00	0.00	0.06	0.15	0.14

Notes: Business-cycle frequencies correspond to cyclical components with periods between 6 and 32 quarters. The variance decomposition is obtained from the spectral representation of the DSGE model. An inverse first-difference filter is applied to GDP, consumption, and investment to recover their level components. The spectral density is computed from the model's state-space representation using 500 frequency bins covering the range of business-cycle periodicities. The figure shows that liquidity (safety) shocks account for an important share of business-cycle fluctuations.

Figure F.1: Impulse response function to a tax shock



Notes: The figure shows the impulse response functions to a one-standard-deviation tax shock in the estimated DSGE model. The blue line corresponds to the estimated Taylor rule, while the yellow line corresponds to the rule with no response to the natural rate, $\phi_{R^*} = 0$. The panels display, in order, output, inflation, investment, consumption, the nominal interest rate, public debt, hours worked, marginal cost, and the natural rate of interest.

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